DTU

02225 DRTS: **Distributed Real-Time Systems**

AVB's Worst-Case Response Time Analysis in TSN

Introduction

Real-time systems

In the context of real-time systems, throughout this course we have studied how such systems are characterized by strict timing requirements, and how these requirements can be specified and analyzed.

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Real-time systems → Timing requirements

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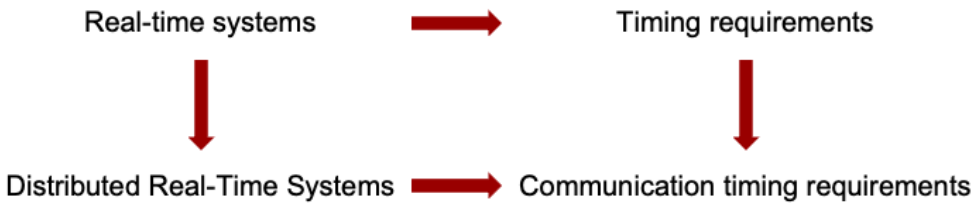
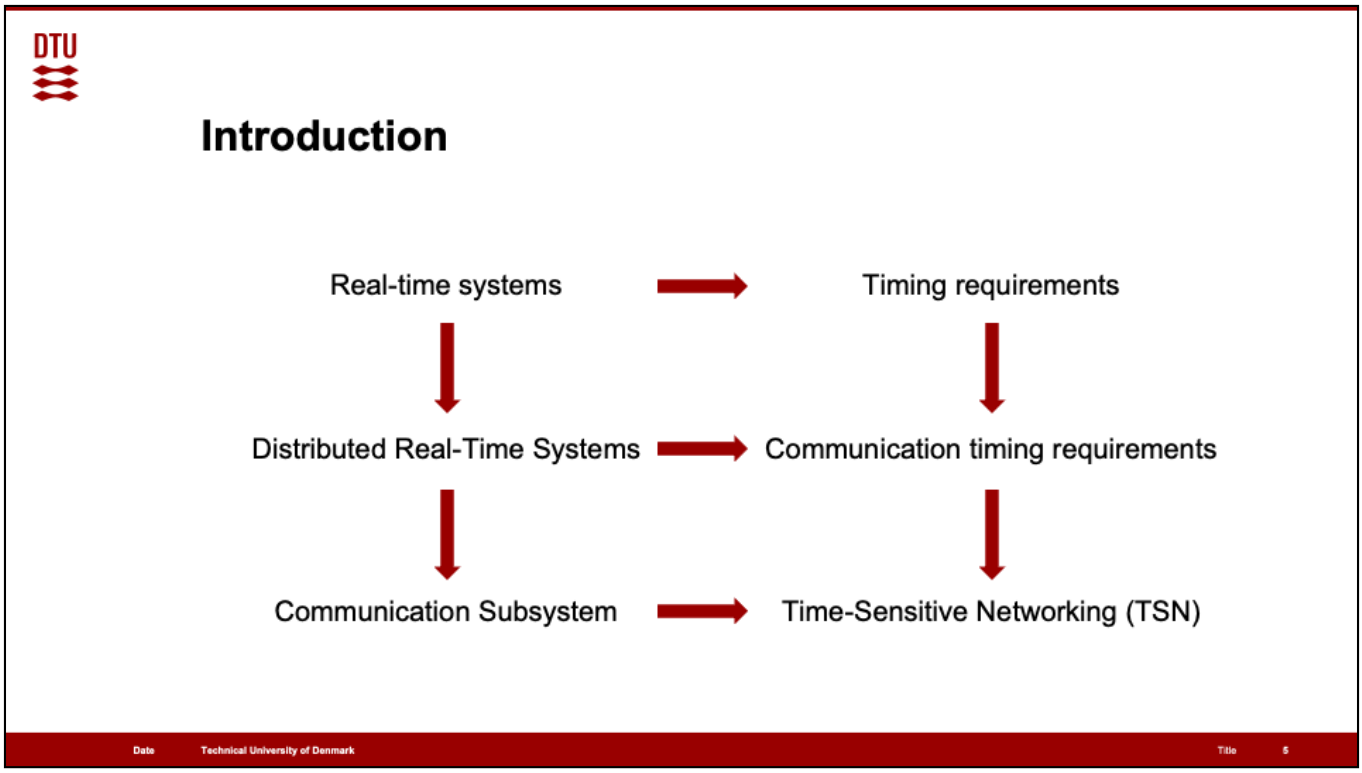
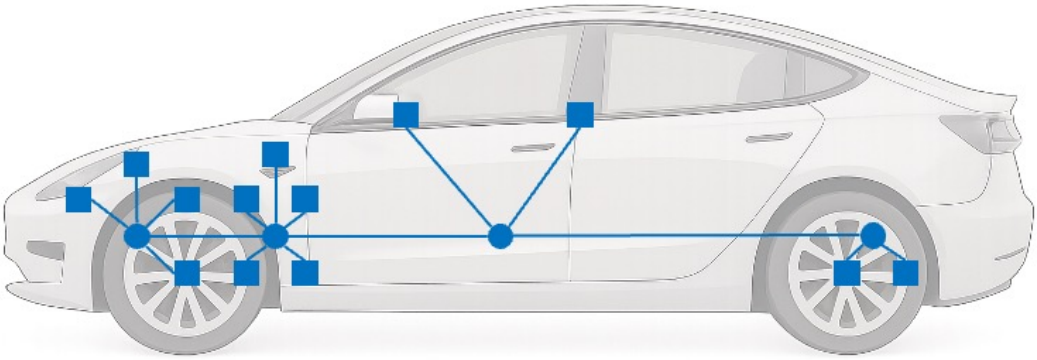


In the case of **distributed real-time systems**, these timing requirements extend beyond computation to include **communication** among distributed components. Consequently, the communication infrastructure itself must be able to provide predictable and bounded timing behavior.

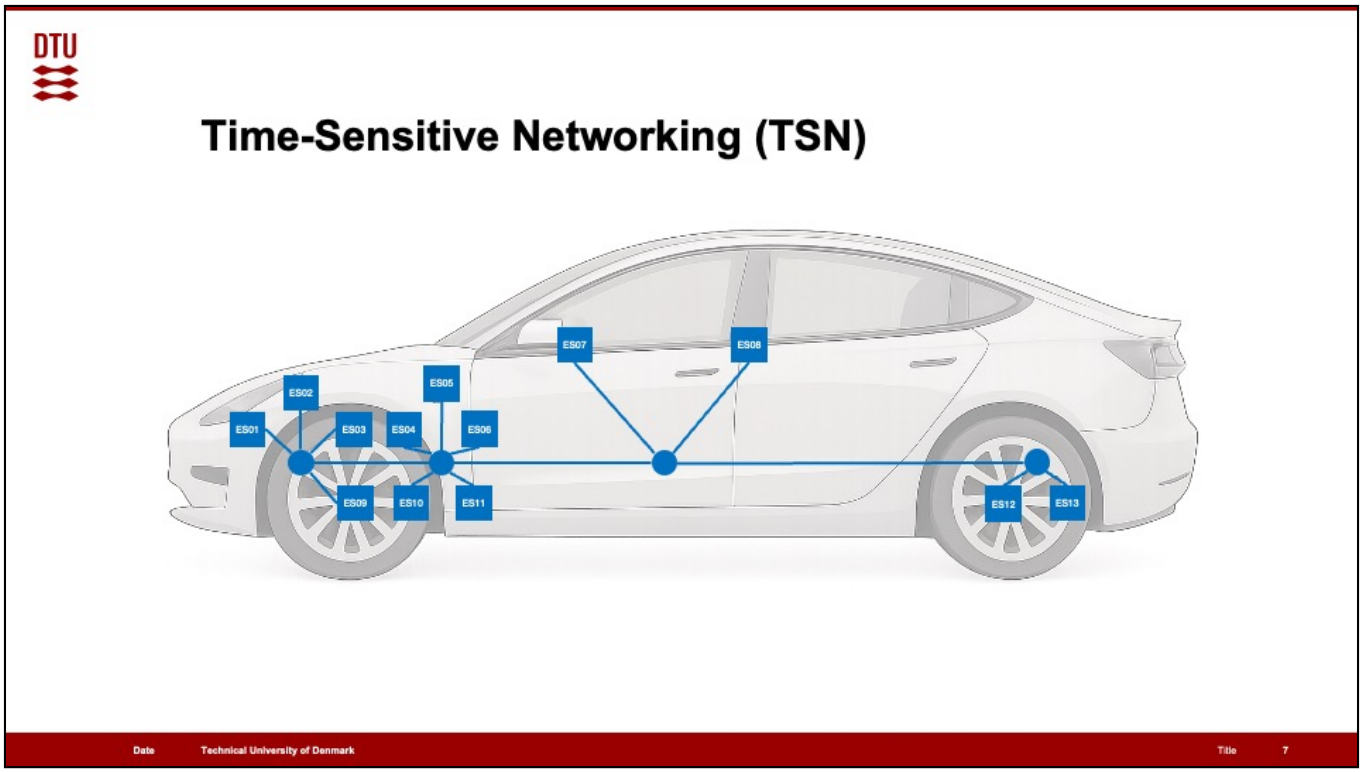


To address this challenge, specialized communication subsystems have been developed that are capable of meeting real-time constraints. One prominent example is **Time-Sensitive Networking (TSN)**, which extends standard Ethernet technologies to support deterministic, real-time communication.

Time-Sensitive Networking (TSN)



In **Time-Sensitive Networking**, as in **switched Ethernet**, systems exchange data packets—referred to as **frames**—over communication **paths** composed of **links** and **switches**.



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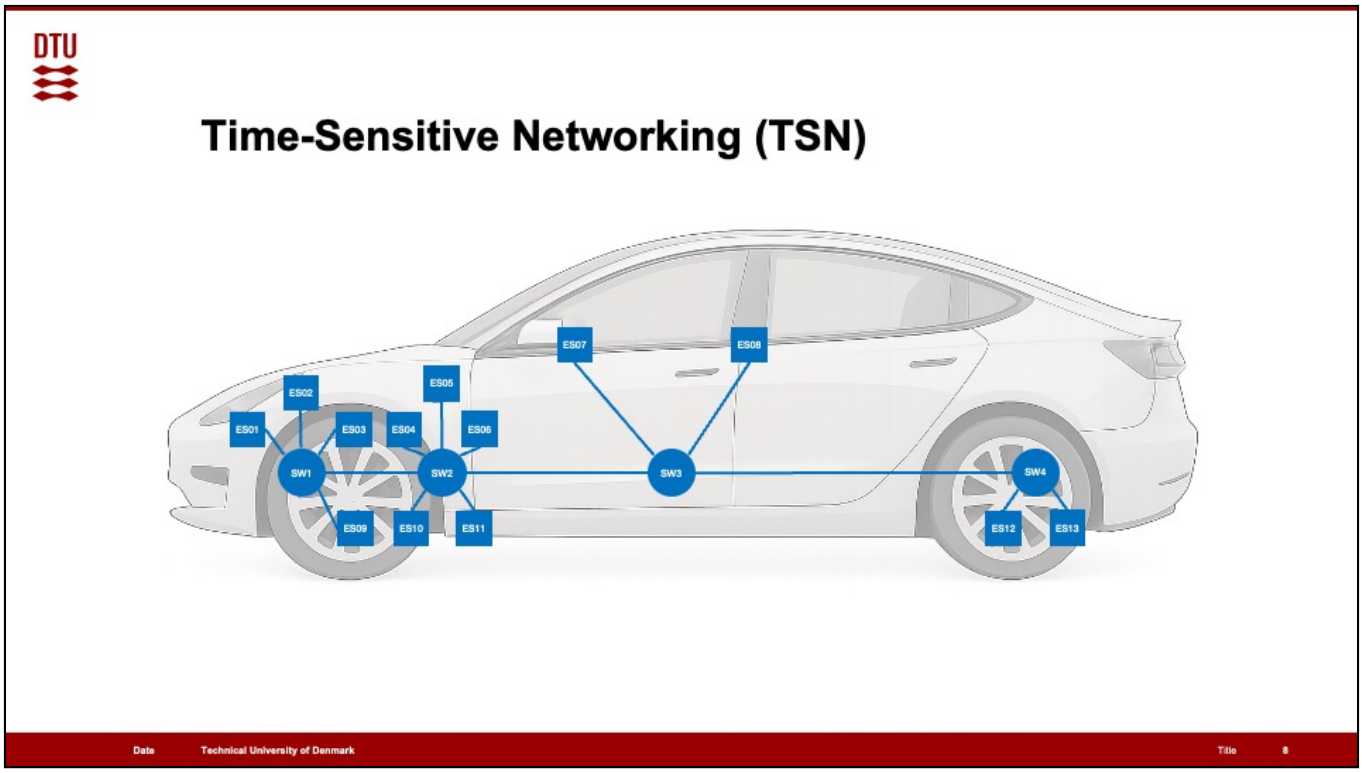

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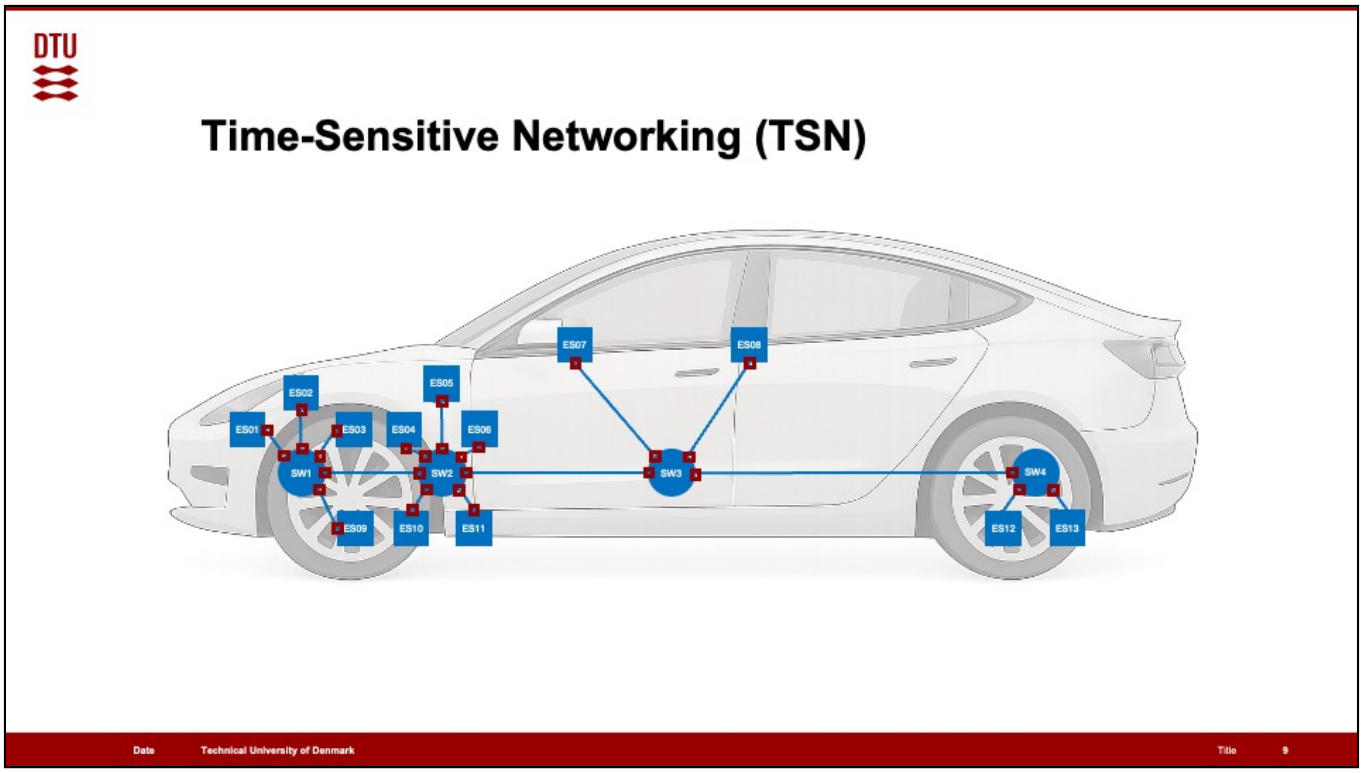
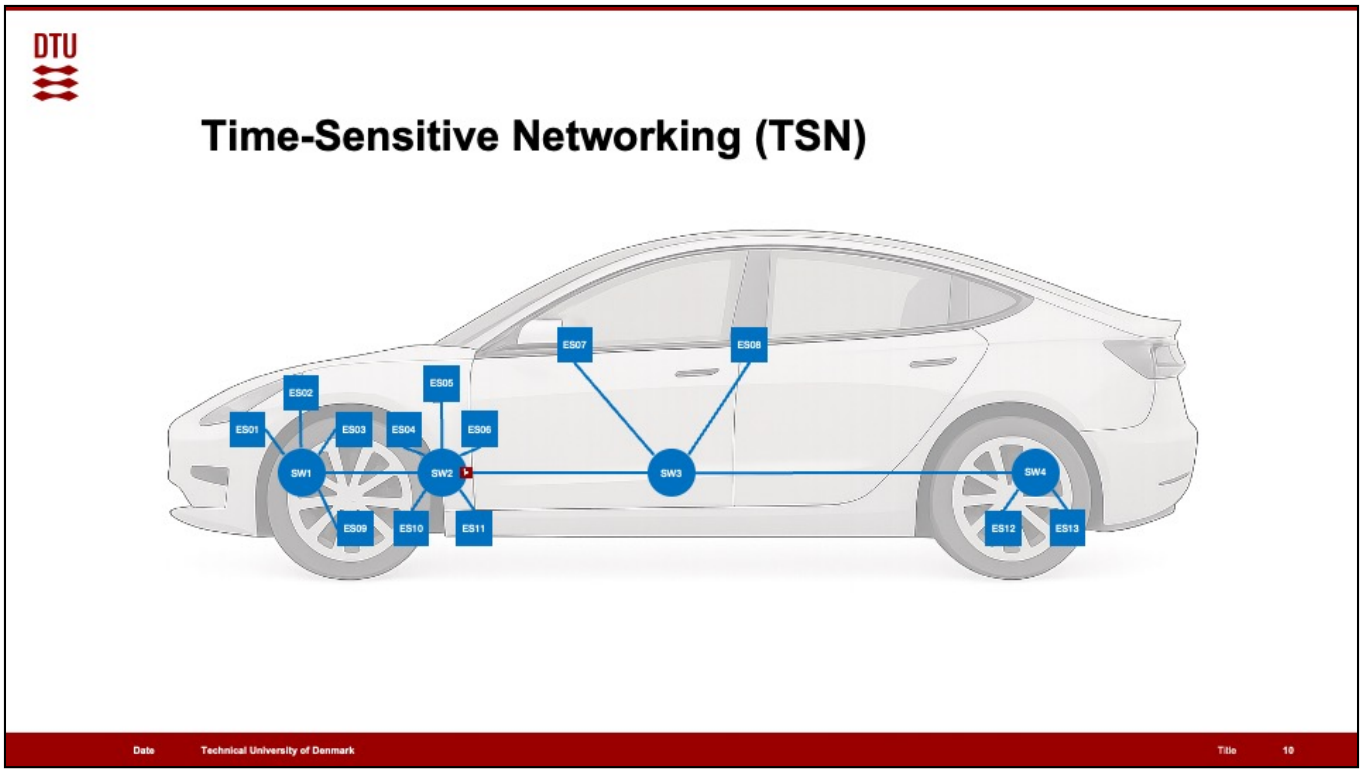


In addition, in order to provide **real-time communication guarantees**, **Time-Sensitive Networking** implements a set of mechanisms at **each output port of every node in the network**.

These mechanisms are specifically designed to control and schedule frame transmissions so as to ensure predictable timing behavior. They are called **traffic shapers** and will be **described and analyzed in detail in the following sections**.



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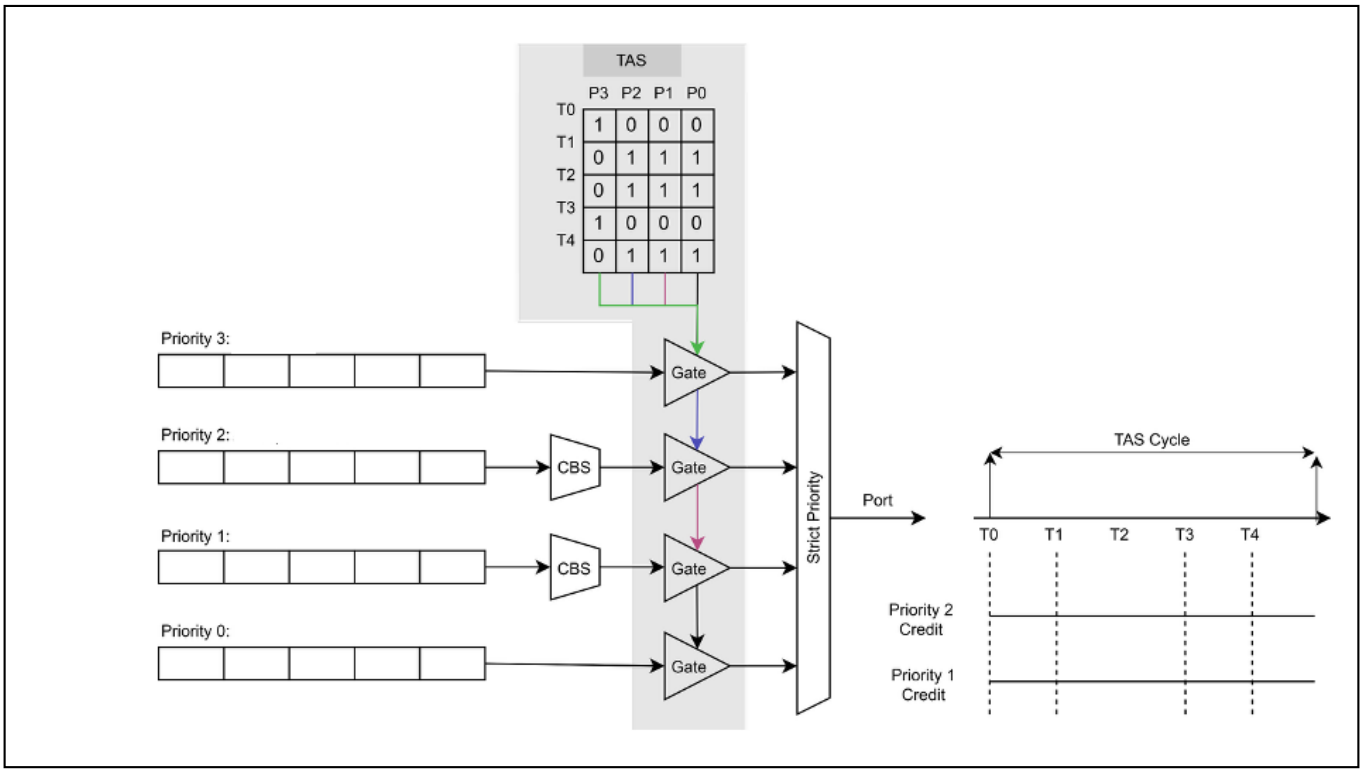

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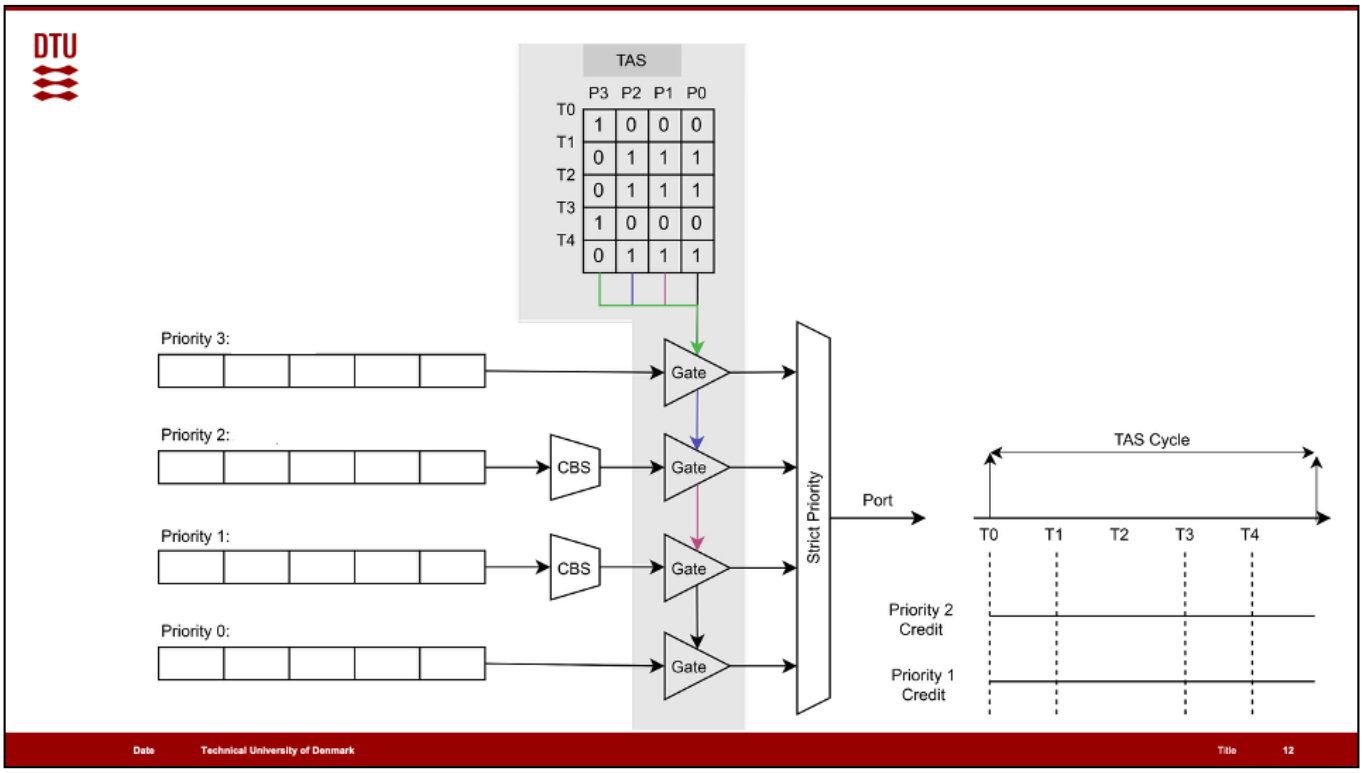
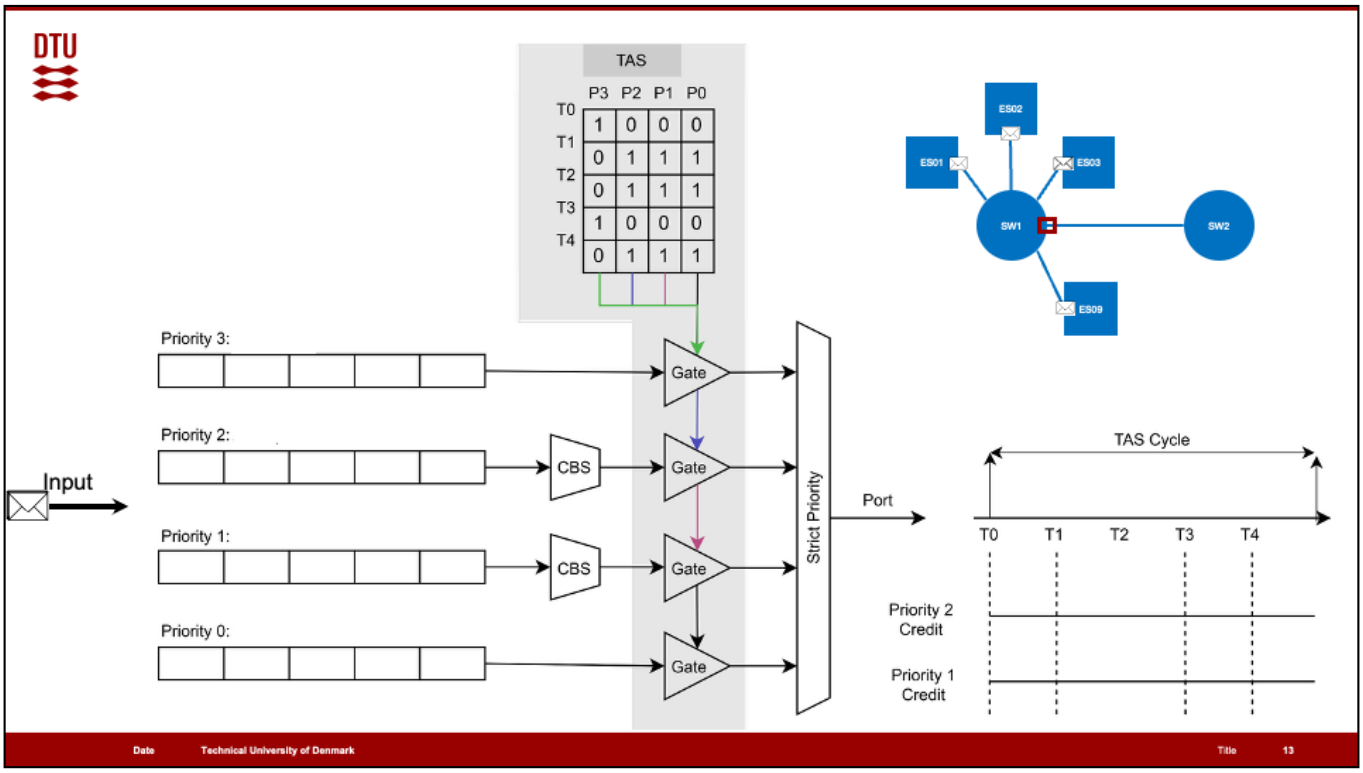
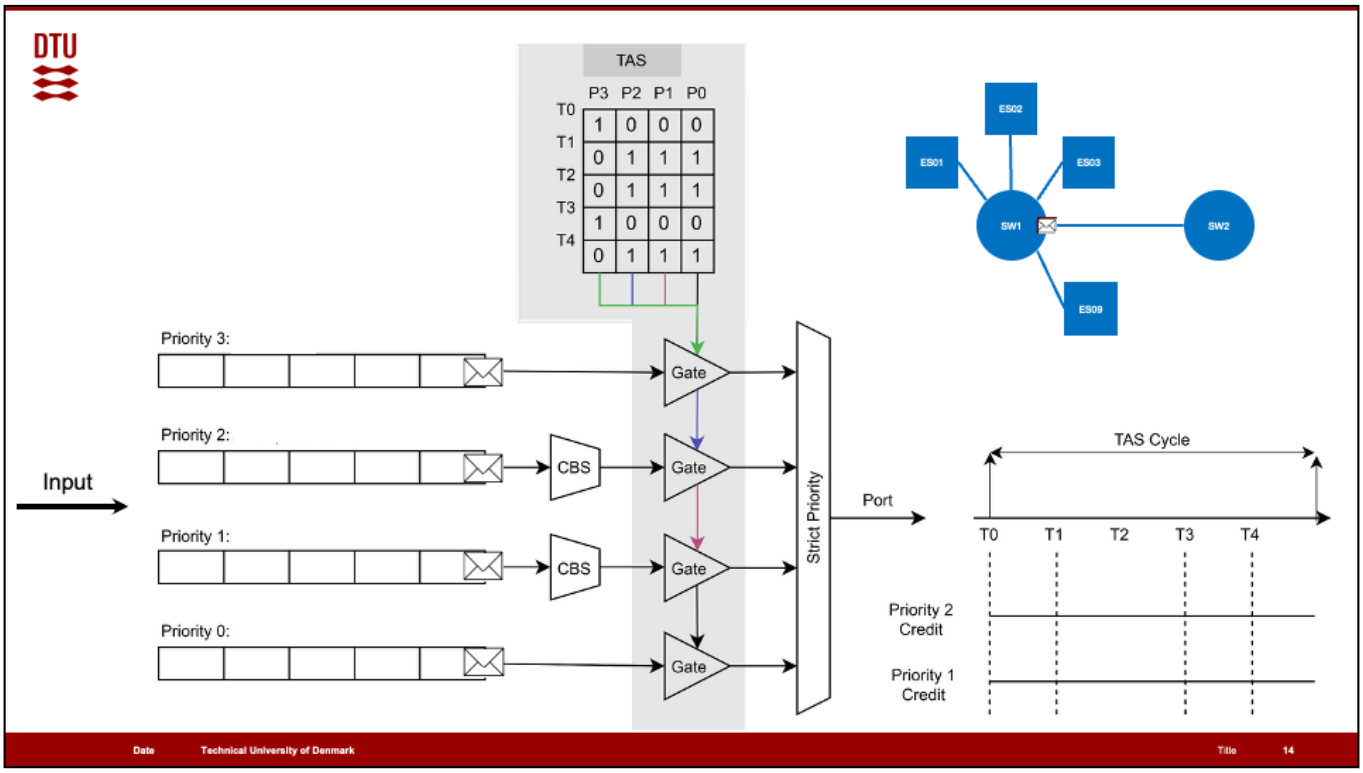


This is how the **traffic shaper structure of an output port** in a **TSN** device is typically represented.

Although the diagram may appear **confusing at first**, do not worry. At this stage, you only need to understand the **overall organization** of the architecture. All the blocks shown in the diagram correspond to the **traffic shaping mechanisms implemented at the output port of each TSN device**.



Traffic enters the diagram **from the left-hand side**, where frames received through the other **input ports** of the device are classified and queued. On the **right-hand side**, the queued traffic is selected and forwarded for transmission onto the outgoing link.



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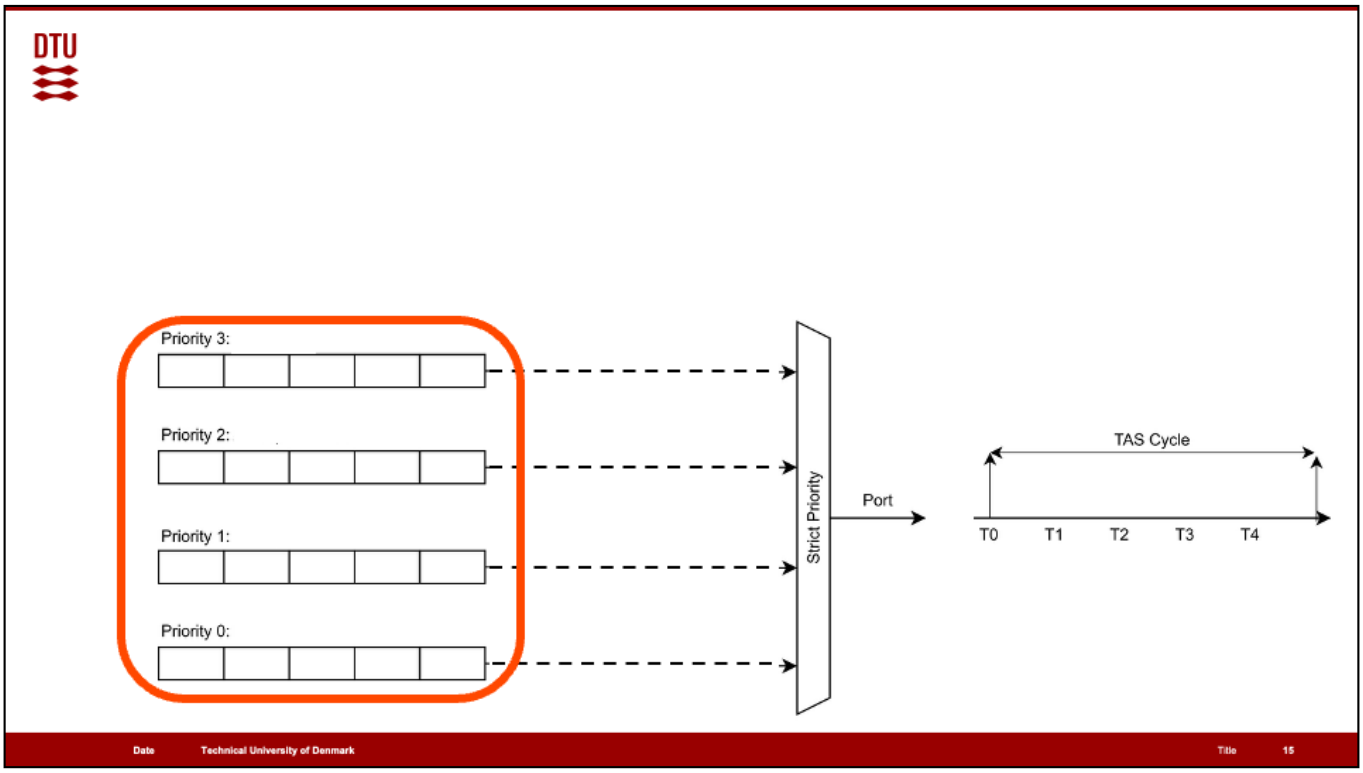
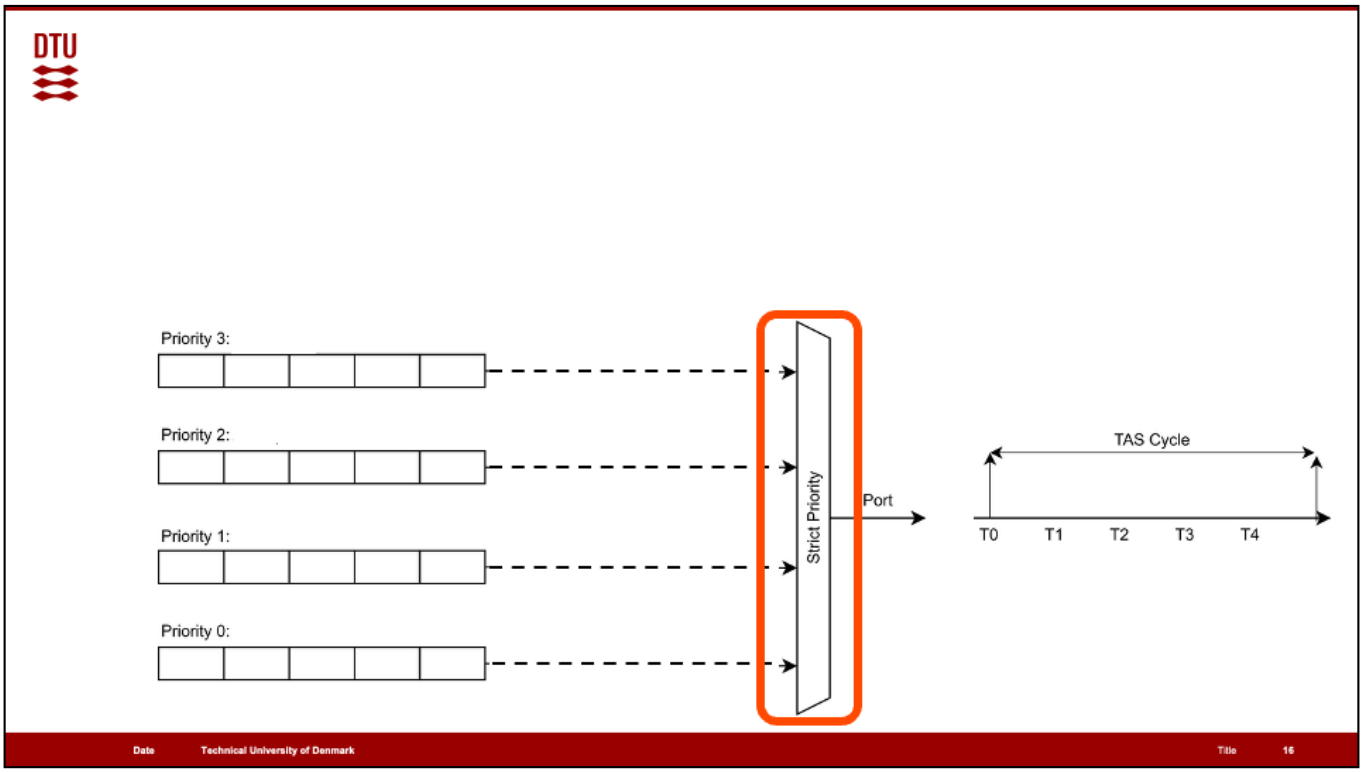
In the following sections, we will progressively analyze each of these blocks and explain their role in providing deterministic, real-time communication guarantees.

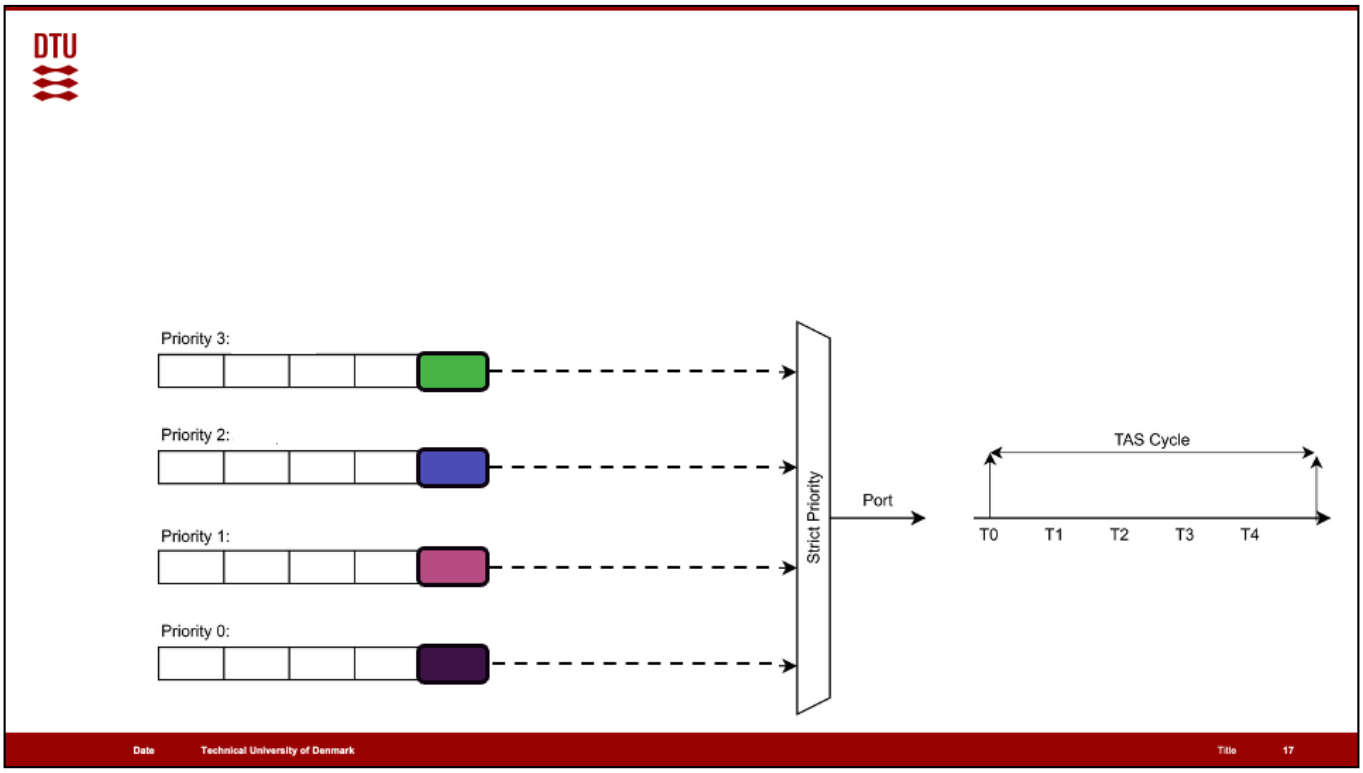


First of all, every **TSN** output port supports **up to eight FIFO queues,** each one associated with a **different priority level.** However, in this example we'll consider only 4 queues for simplicity.

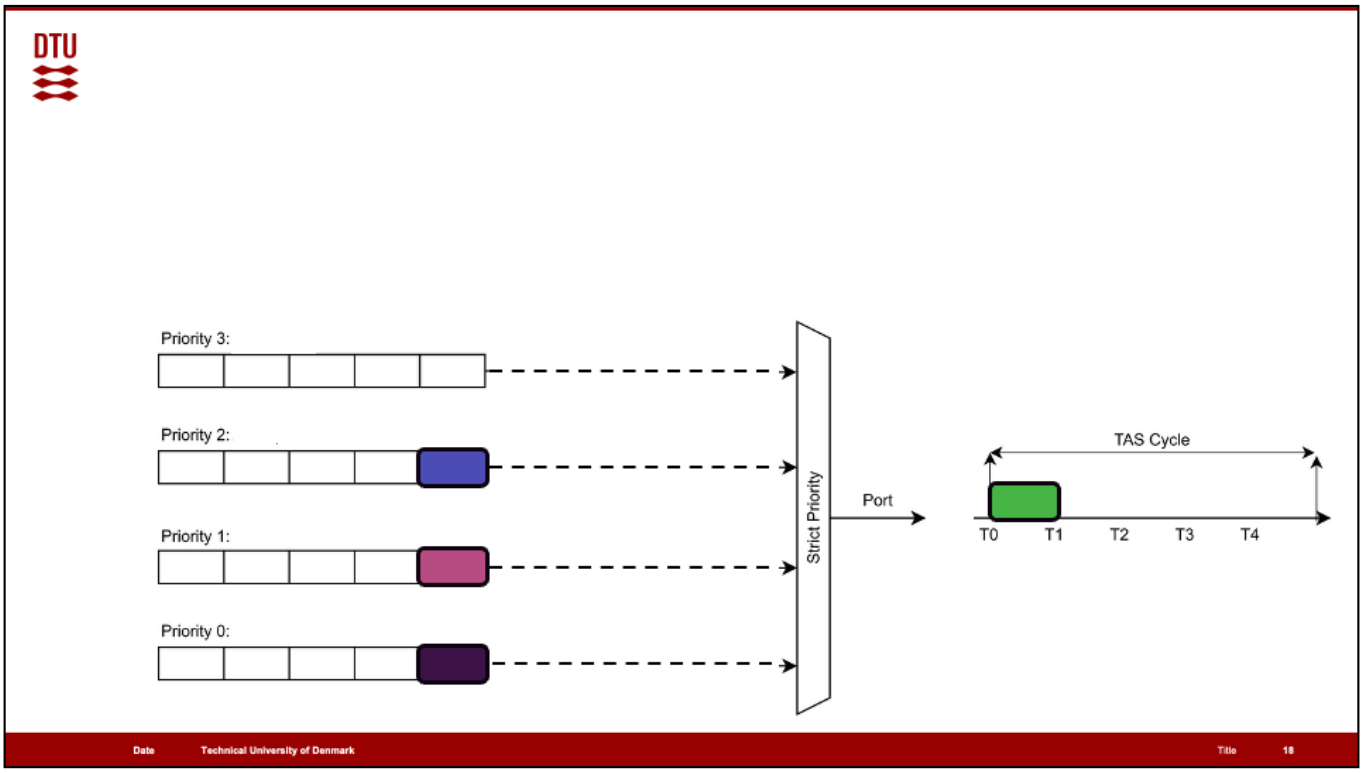


Therefore, the first traffic shaping mechanism is **strict priority scheduling**. The principle is simple: if two or more frames are eligible for transmission at the same time, the frame belonging to the **highest-priority queue** is selected and transmitted first.

This mechanism ensures that high-priority traffic experiences minimal delay, although—if used alone—it may lead to starvation of lower-priority traffic, a limitation that will be addressed by additional TSN mechanisms introduced later.


Priority 3:
Priority 2:
Priority 1:
Priority 0:
Strict Priority
Port
TAS Cycle
T0
T1
T2
T3
T4
Date
Technical University of Denmark
Title
17

EXAMPLE


Priority 3:
Priority 2:
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Technical University of Denmark
Title
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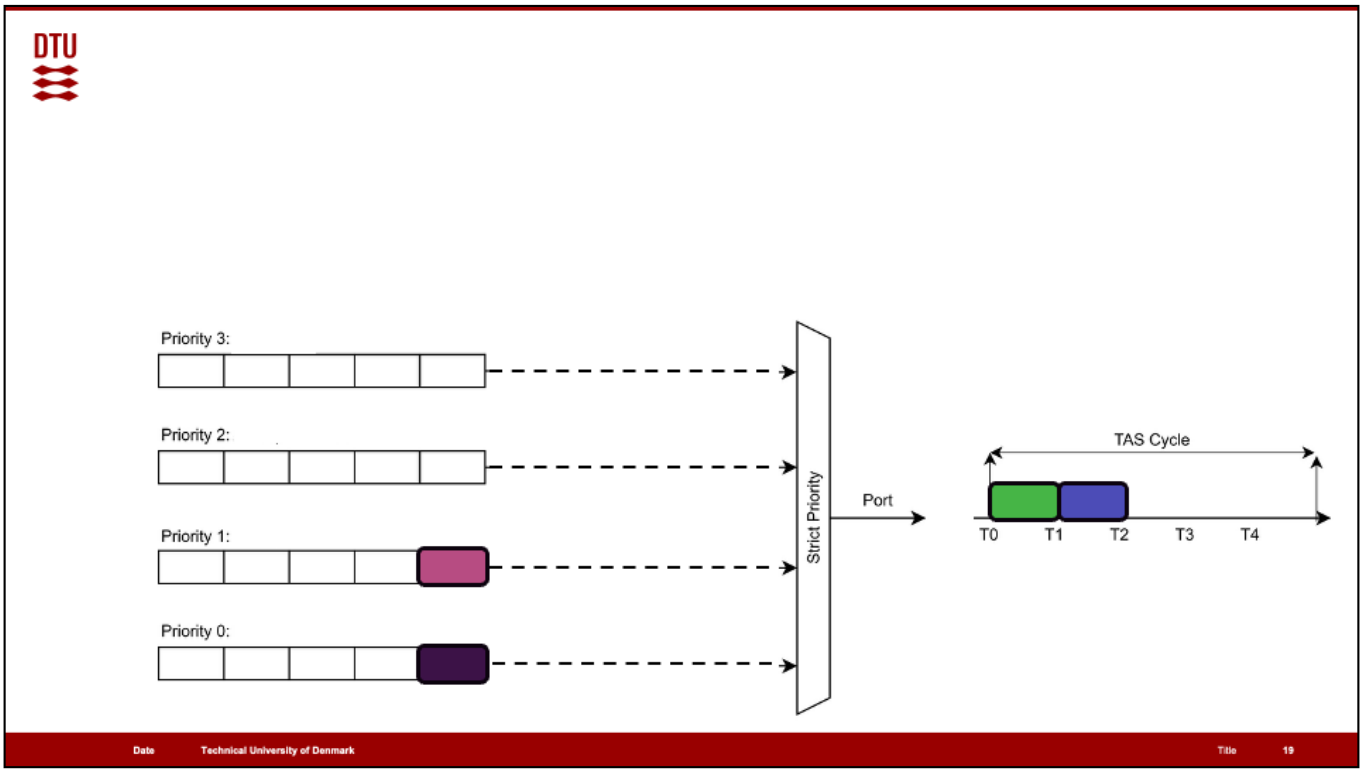
EXAMPLE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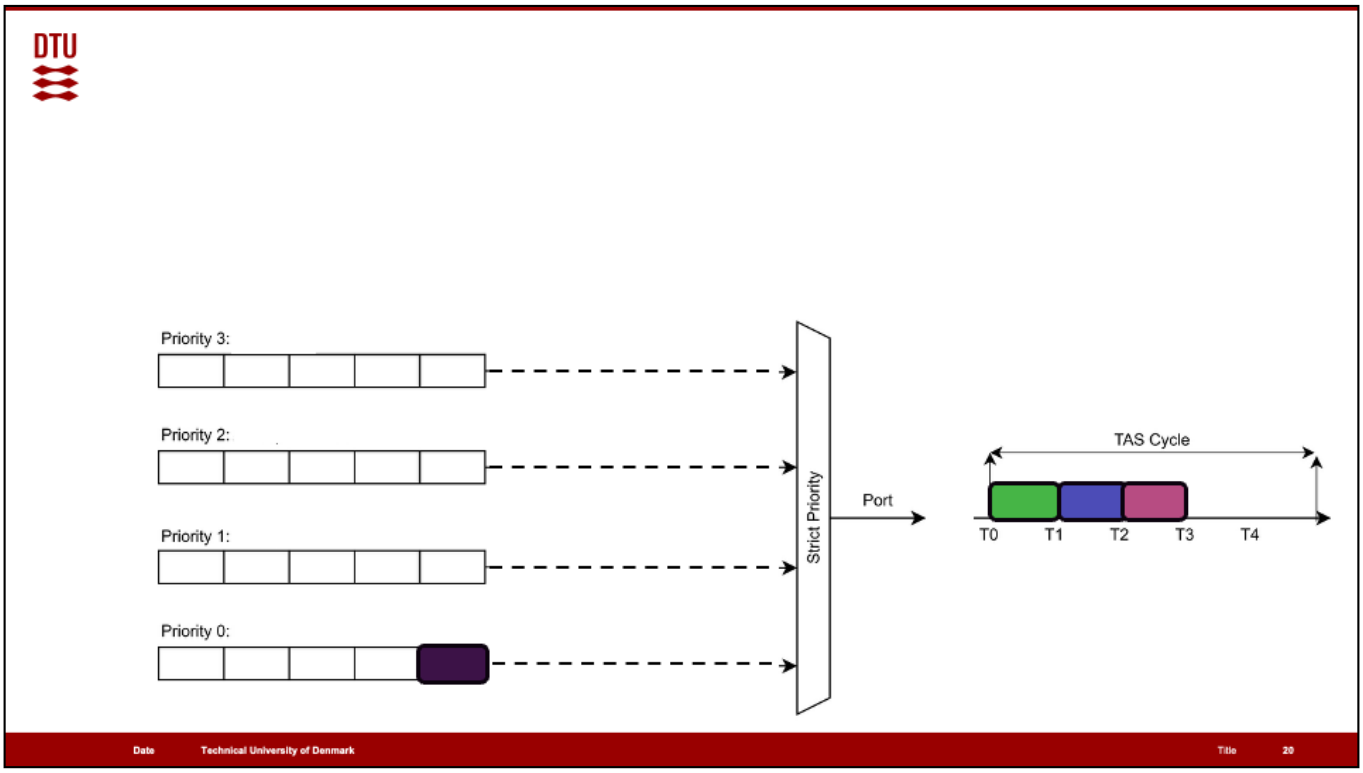
Date
Technical University of Denmark
Title
19

EXAMPLE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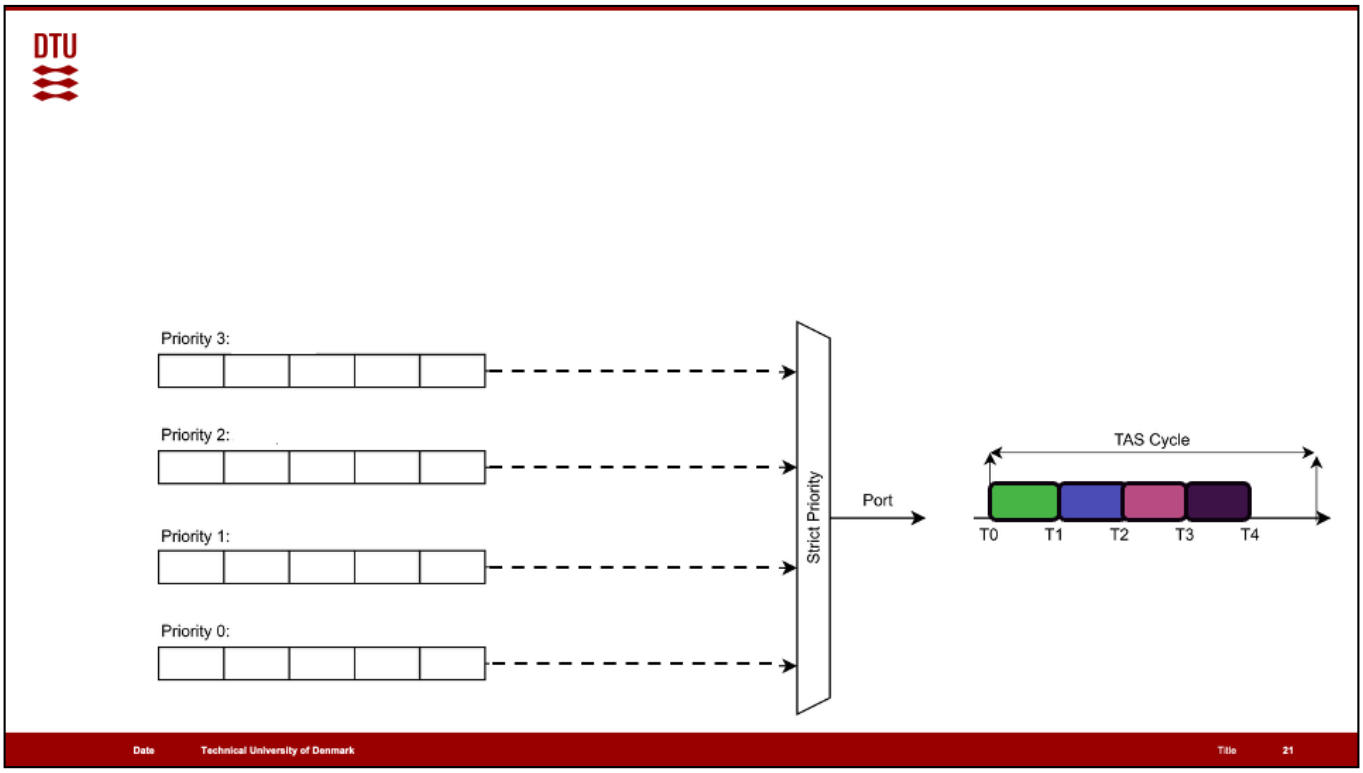
Priority 3:
Priority 2:
Priority 1:
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Port
TAS Cycle
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T1
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T3
T4
Date
Technical University of Denmark
Title
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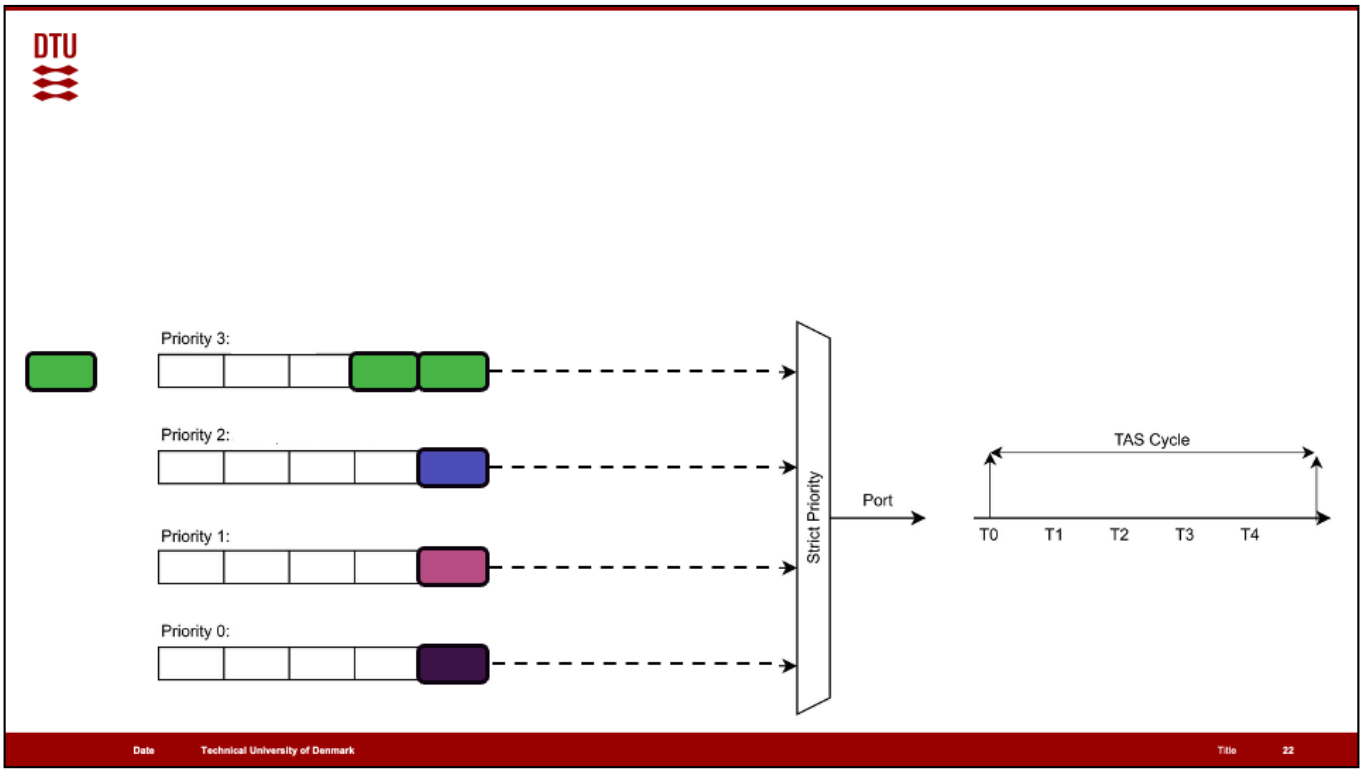
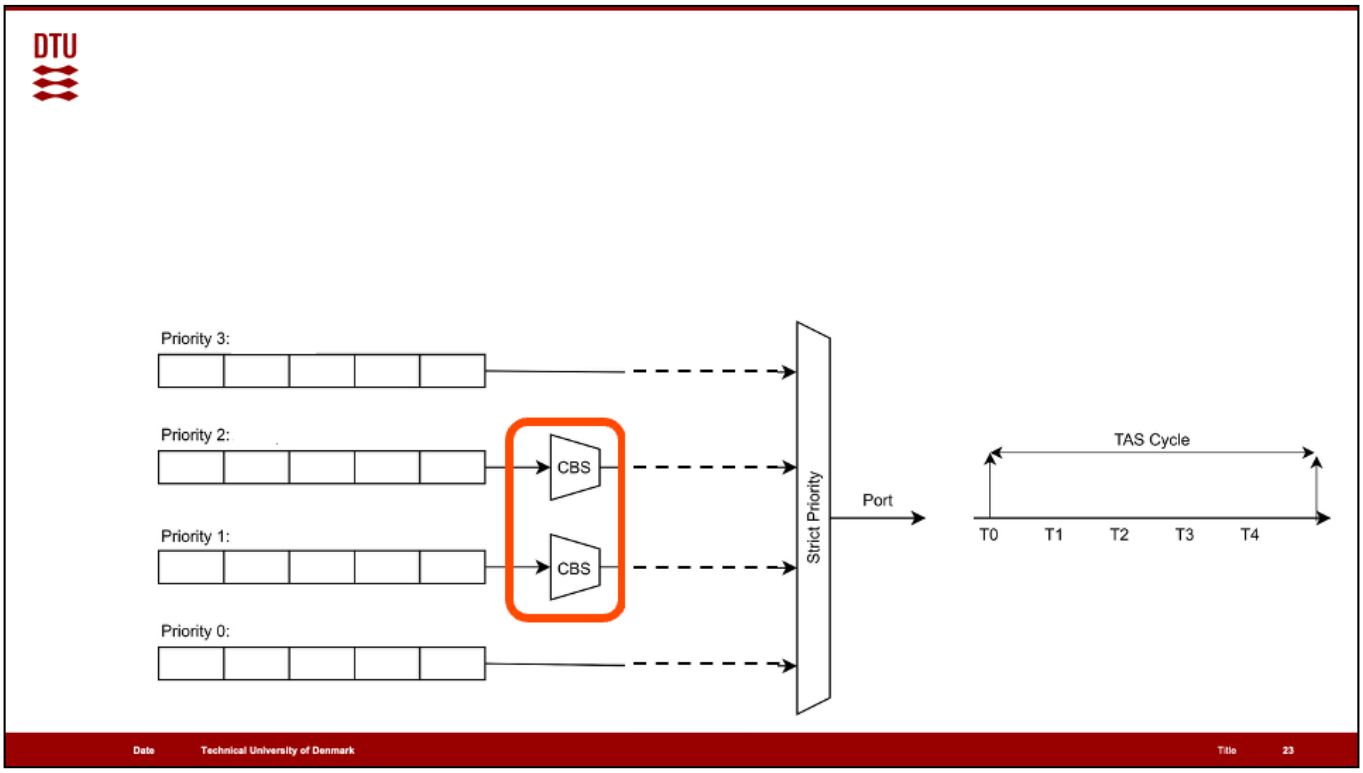
EXAMPLE



EXAMPLE OF STARVATION

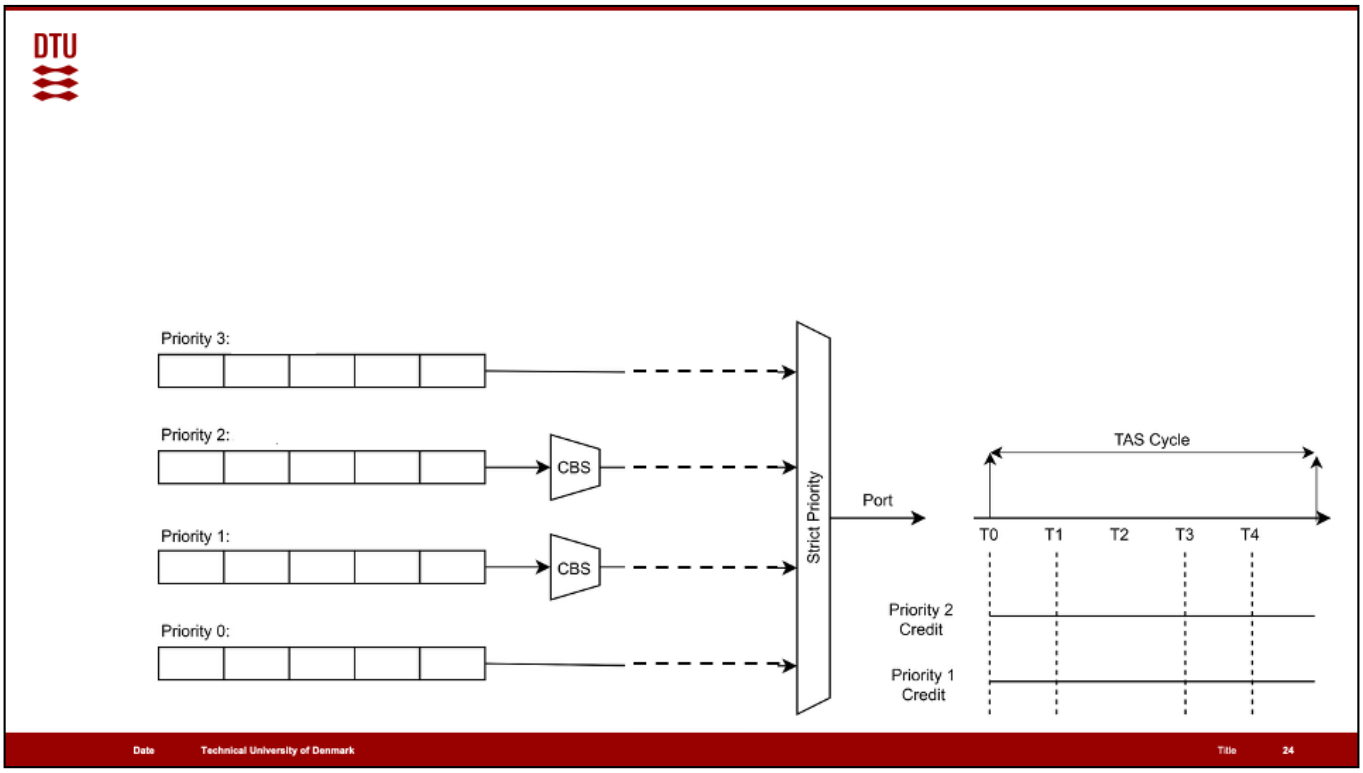


To prevent **lower-priority starvation** while still maintaining **bounded delays for high-priority traffic classes**, **TSN** also includes the **Credit-Based Shaper (CBS)**.

This mechanism assigns a **credit value** to selected priority classes, such that frames from these classes are eligible for transmission **only when their credit is zero or positive**. The credit starts at **zero** and evolves dynamically over time.

When a queue is transmitting, its credit is **consumed at a constant rate**. Conversely, the credit **increases** when there is traffic waiting in the queue but the queue is not transmitting, or when the credit is negative. If the credit becomes positive and there is **no traffic waiting** in the queue, the credit is immediately reset to **zero**.

By regulating transmission eligibility through this credit mechanism, CBS ensures fairness among traffic classes while still providing **deterministic and bounded latency guarantees** for time-critical traffic.


Priority 3:
Priority 2:
Priority 1:
Priority 0:
CBS
CBS
Strict Priority
Port
TAS Cycle
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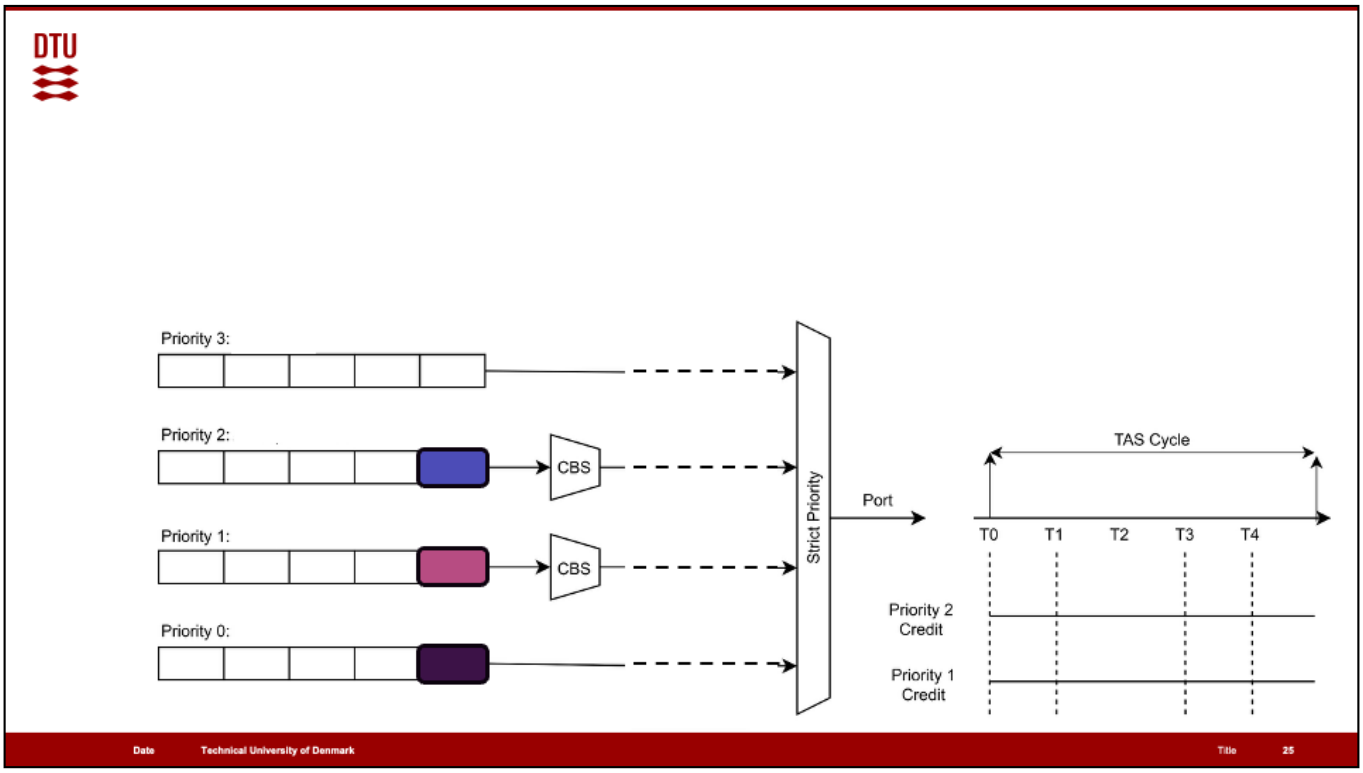
Priority 2
Credit
Priority 1
Credit
Date
Technical University of Denmark
Title
24

EXAMPLE


Priority 3:
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CBS
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Port
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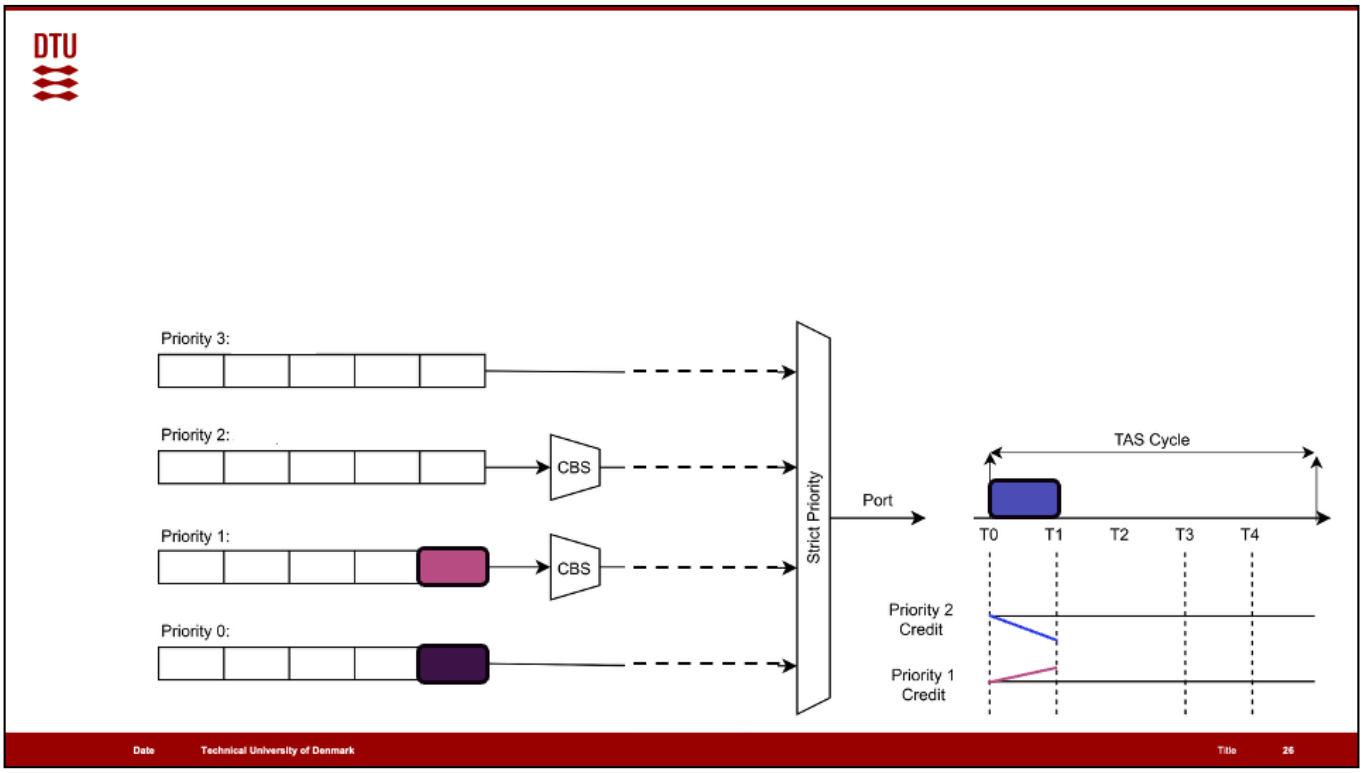
Credit
Date
Technical University of Denmark
Title
25

EXAMPLE


Priority 3:
Priority 2:
Priority 1:
Priority 0:
CBS
CBS
Strict Priority
Port
TAS Cycle
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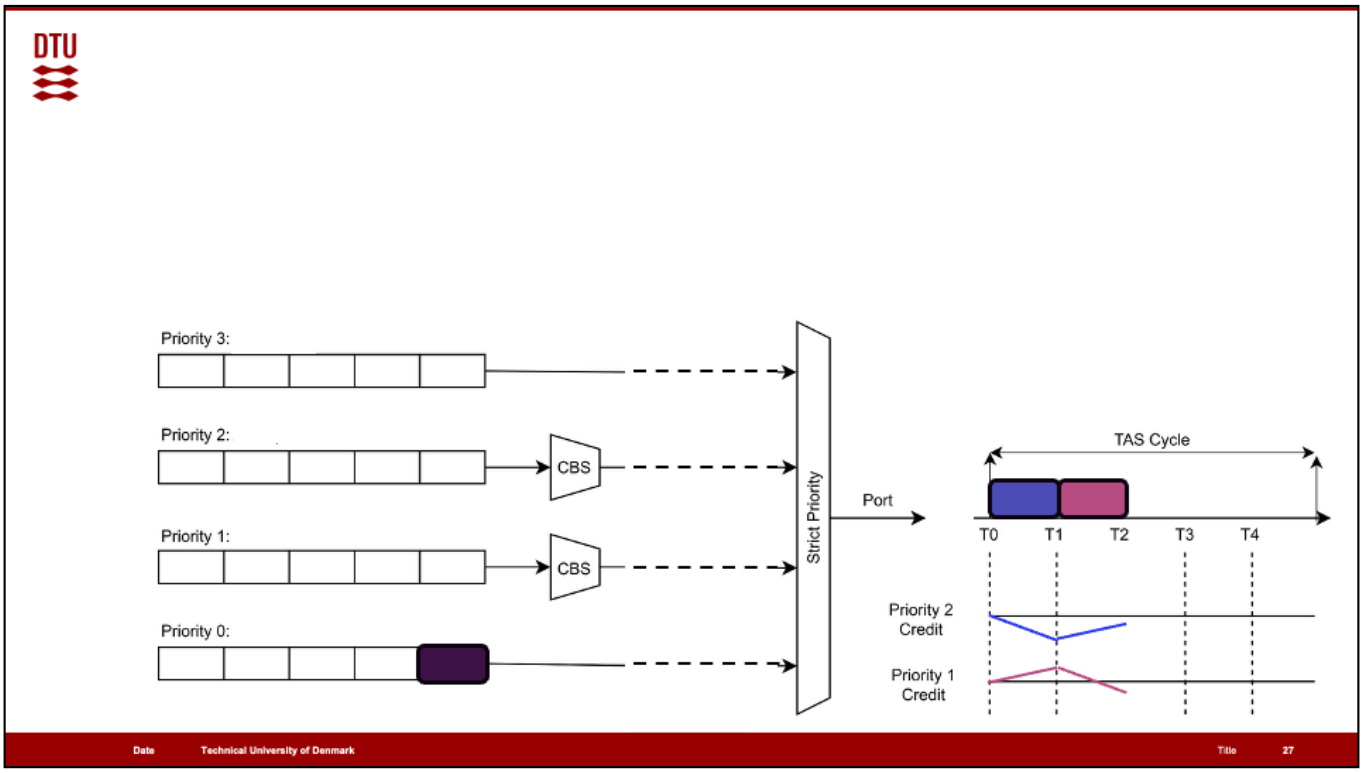
Priority 2
Credit
Priority 1
Credit
Date
Technical University of Denmark
Title
26

EXAMPLE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Strict Priority
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Priority 2
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Priority 1
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Title
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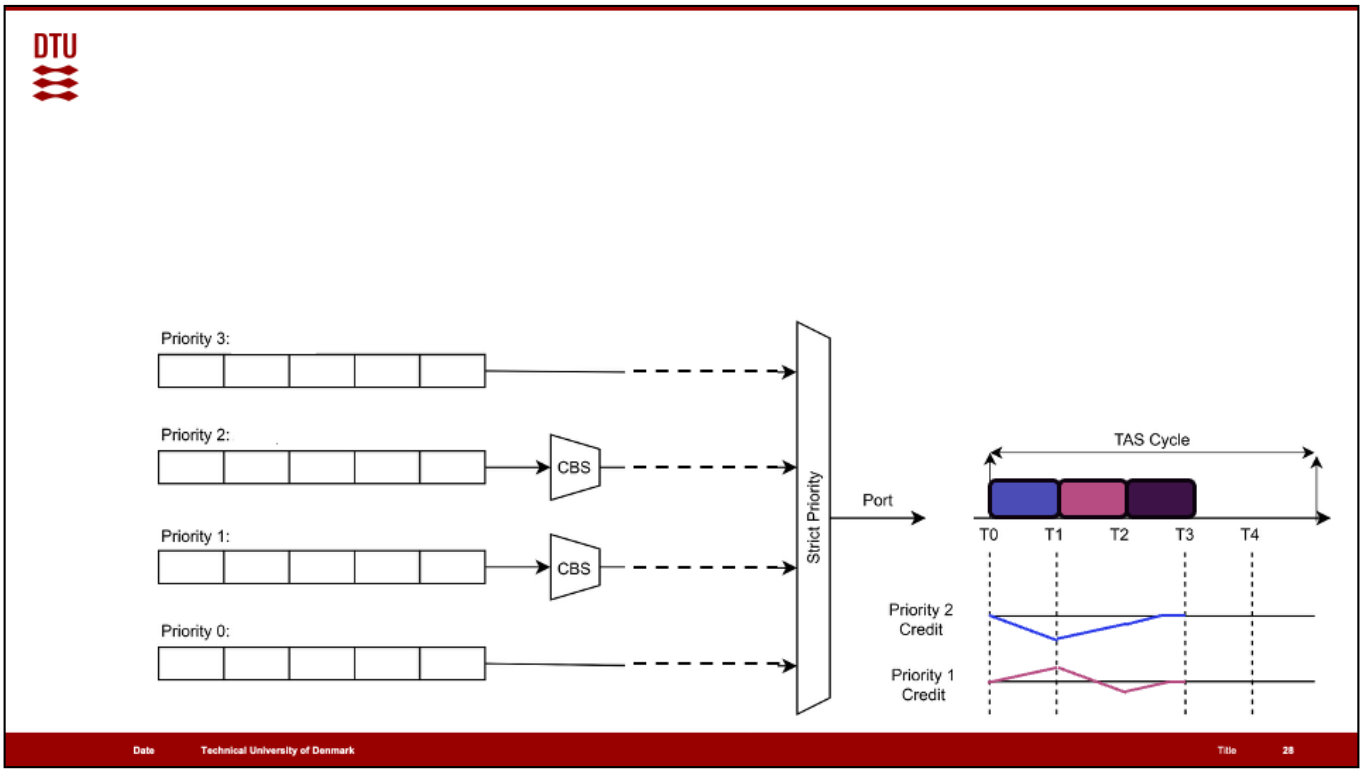
Priority 3:
Priority 2:
Priority 1:
Priority 0:
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CBS
Strict Priority
Port
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28

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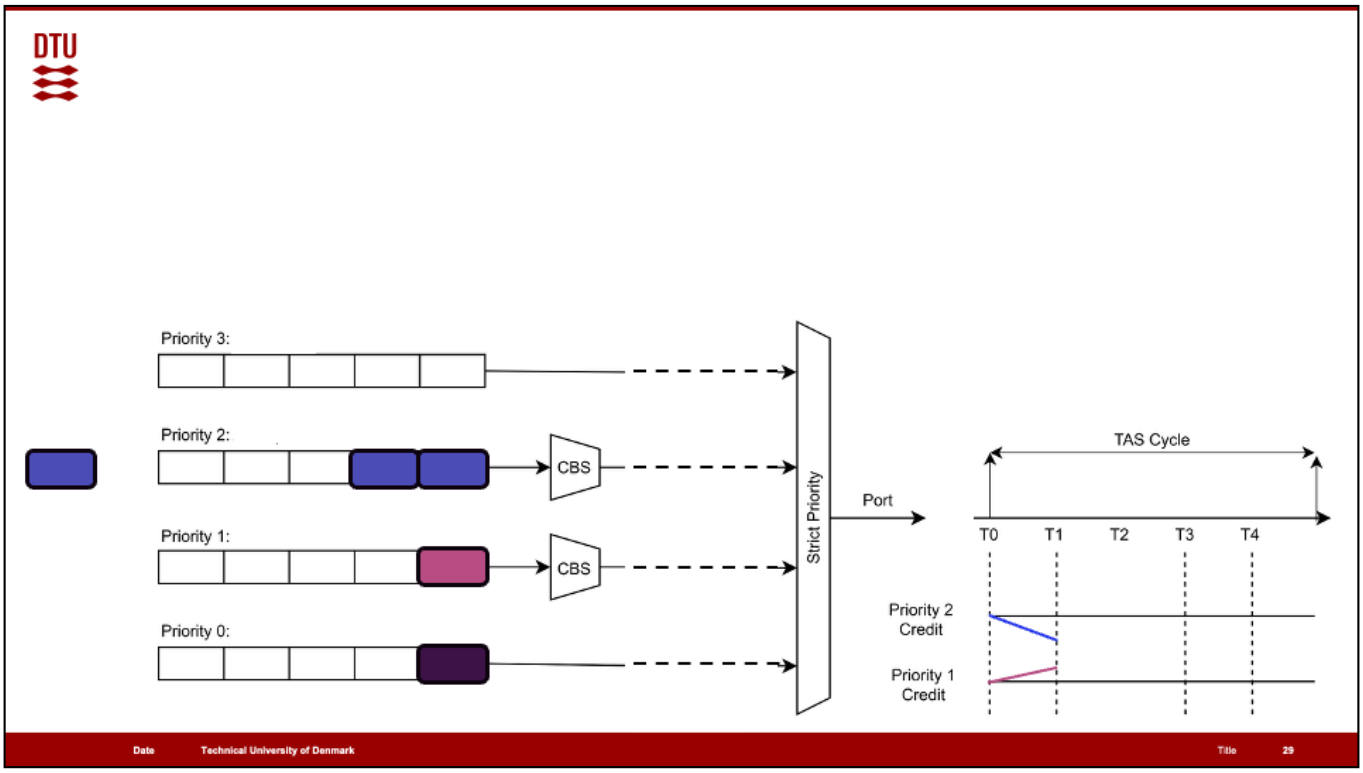
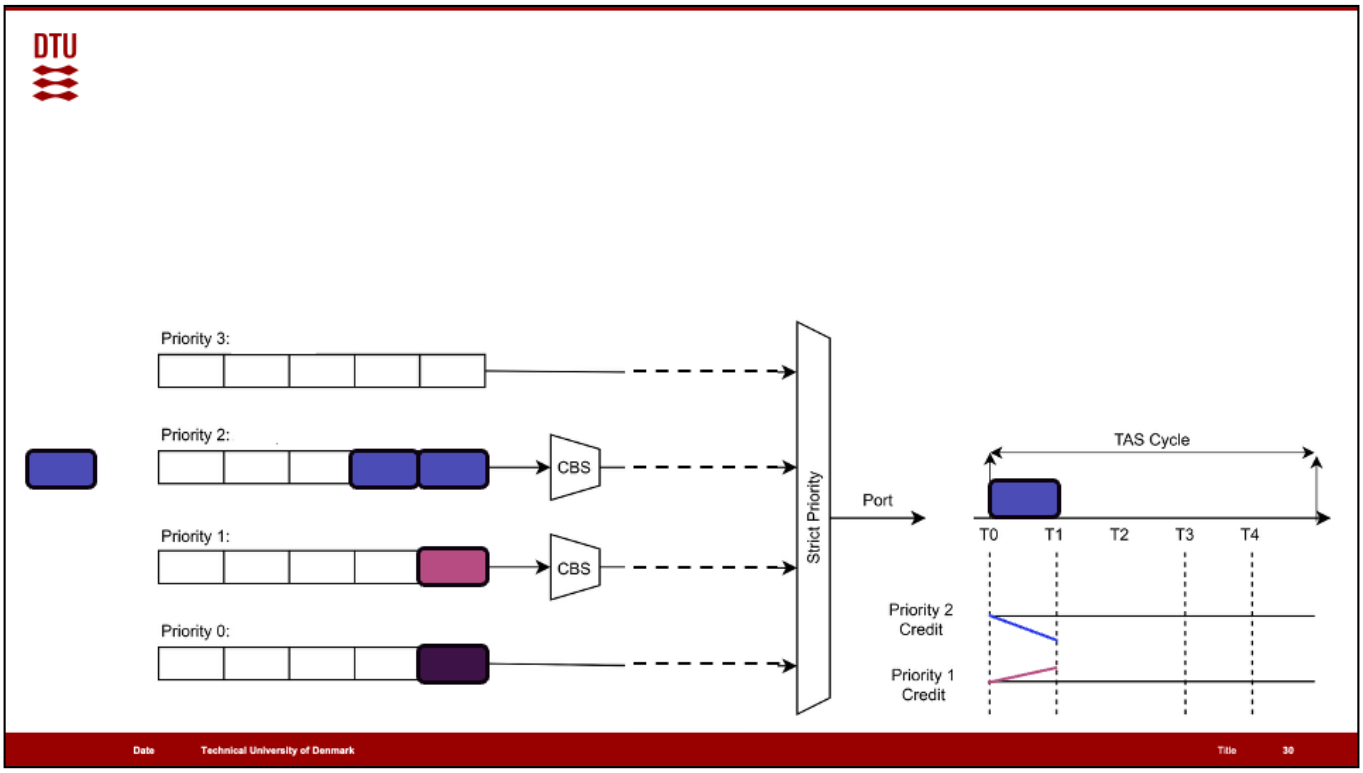


EXAMPLE OF STARVATION



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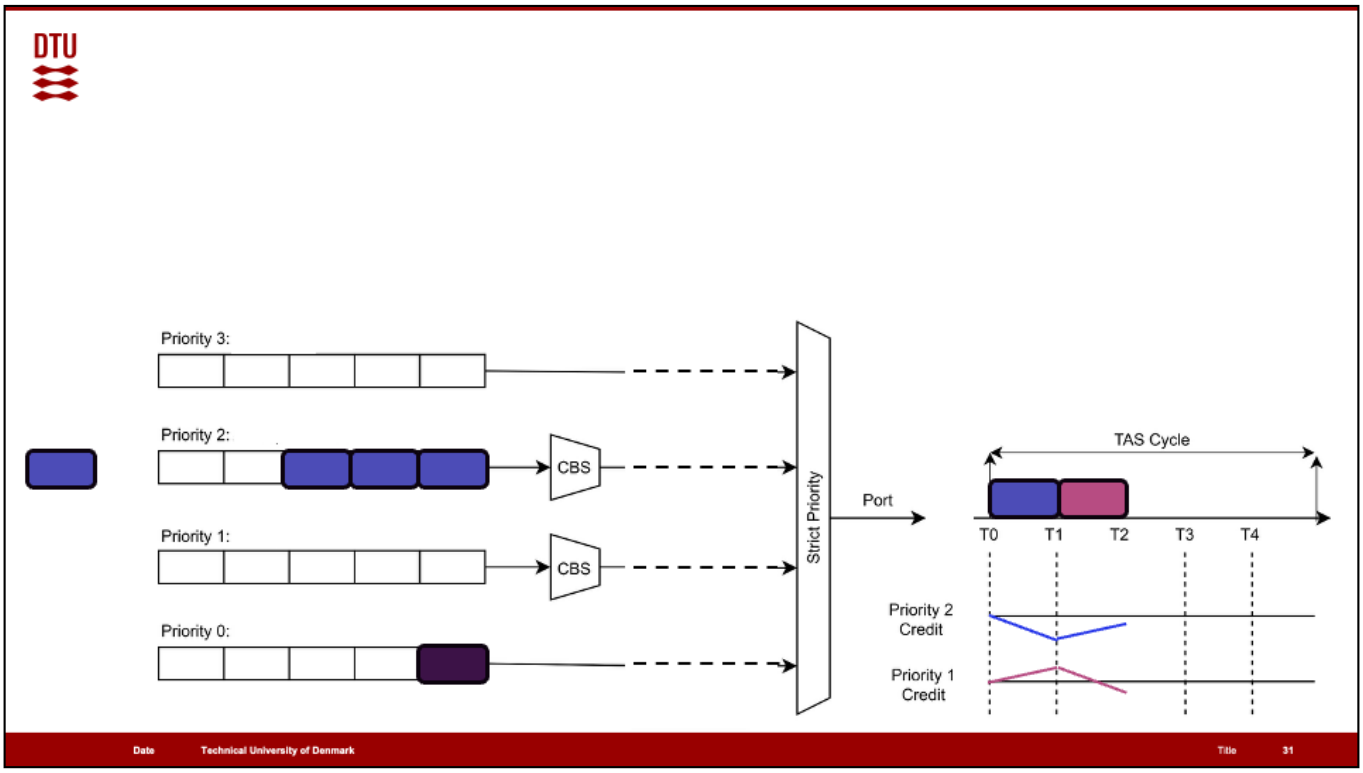

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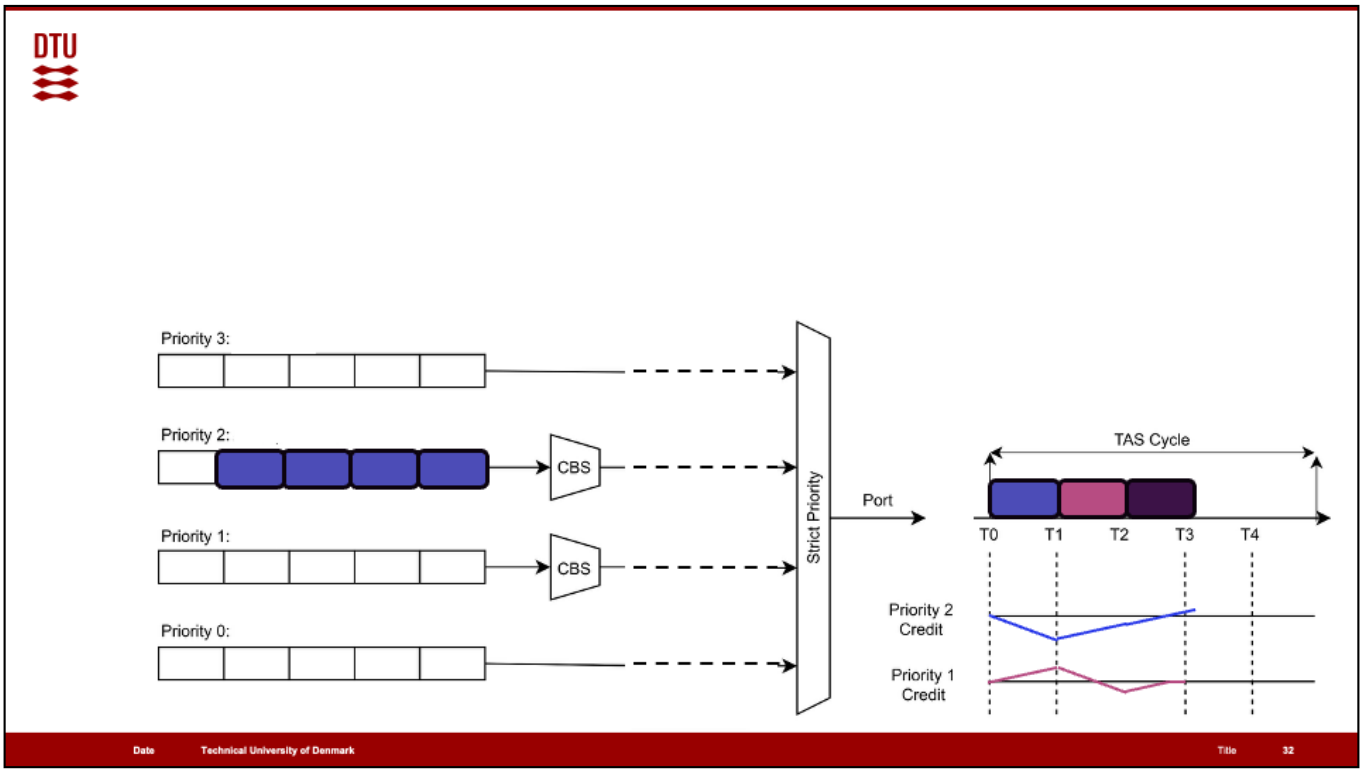

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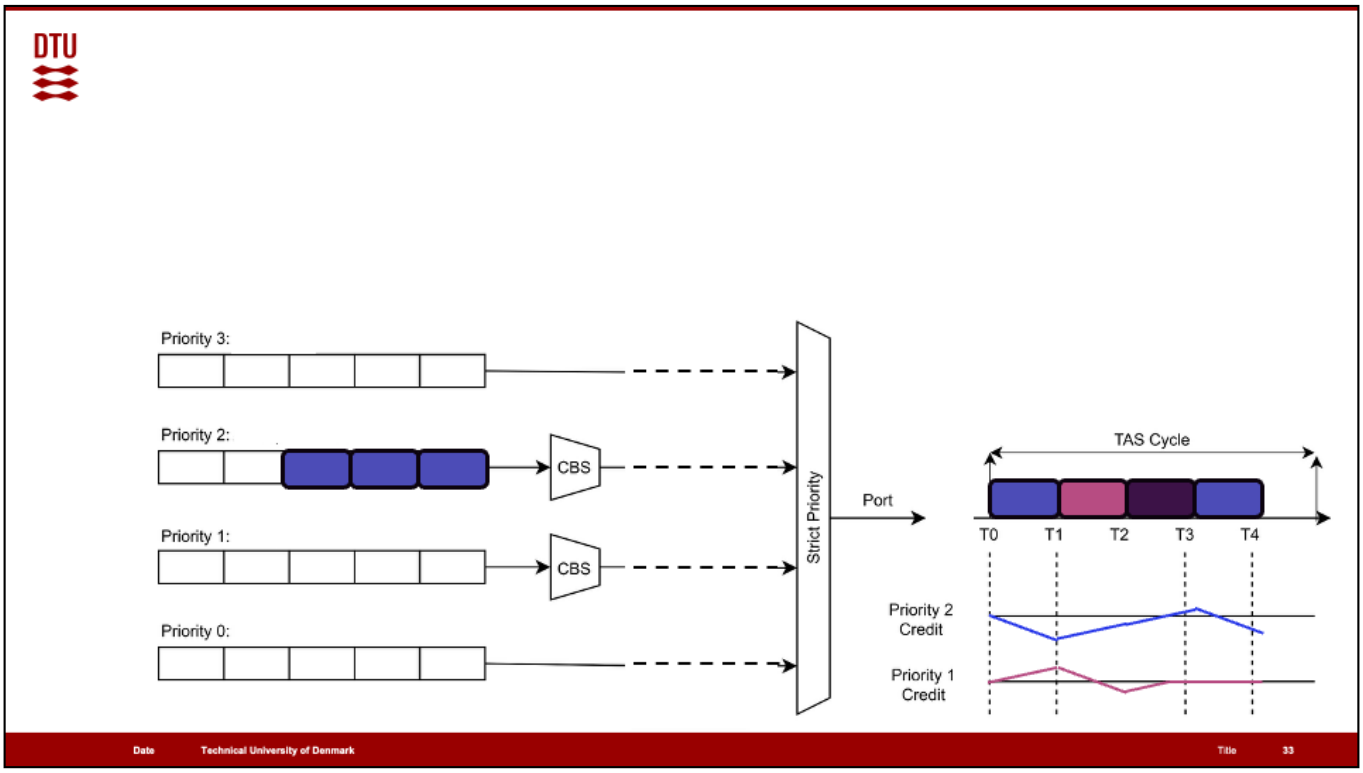

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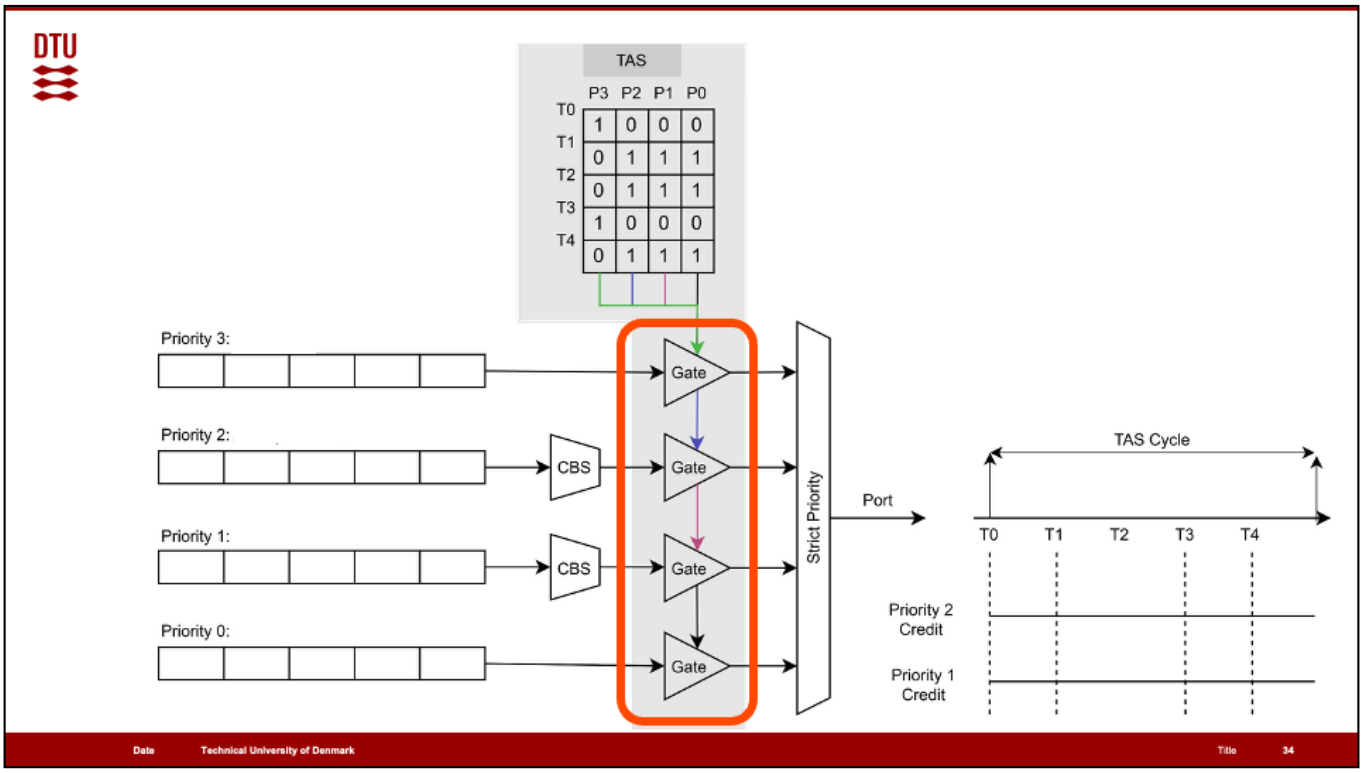
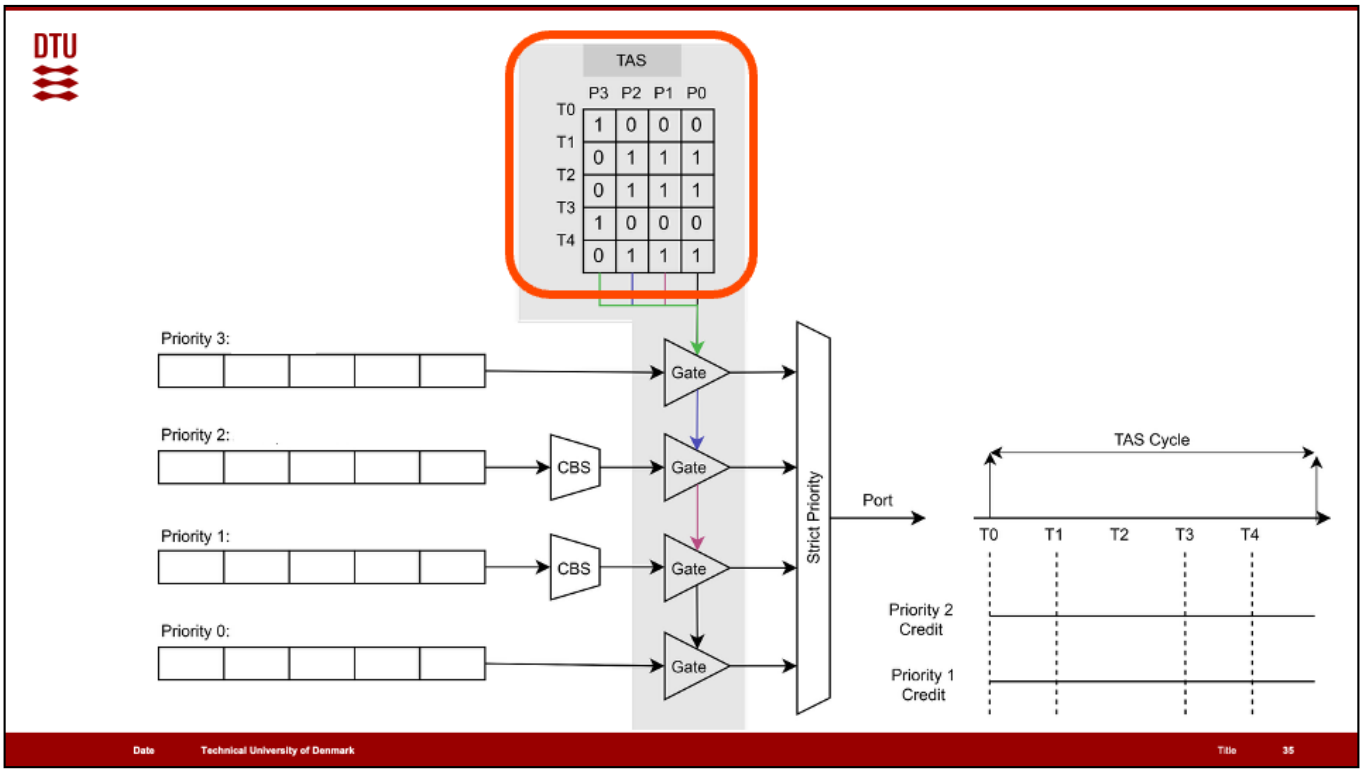


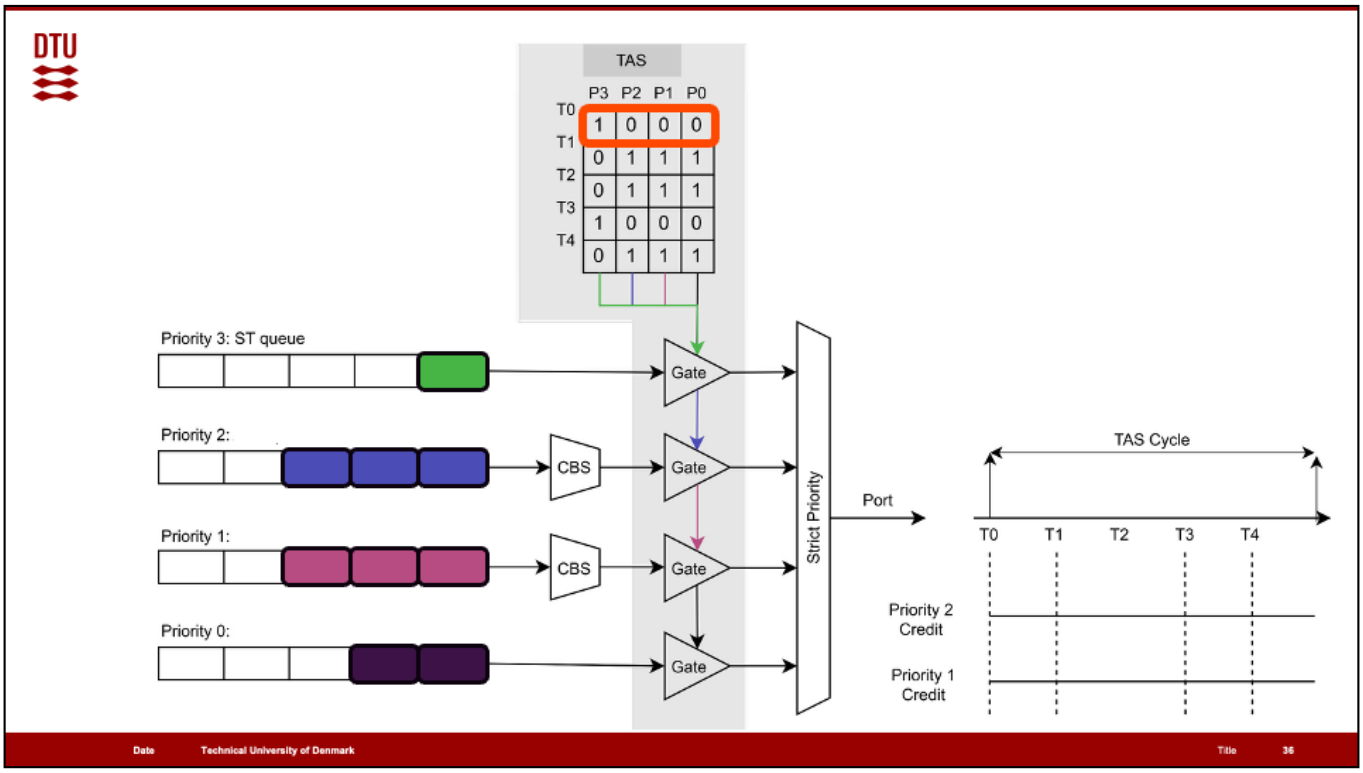
Finally, for traffic with **very strict timing requirements**—which require not only **bounded delay** but also **bounded jitter**—TSN implements a mechanism that allows these queues to be **isolated and scheduled in time**, known as the **Time-Aware Shaper (TAS)**.

TAS associates each queue with a **gate** that can be either **open** or **closed**. When a gate is open, the corresponding queue is **eligible for transmission**; when it is closed, the queue is **blocked** and cannot transmit frames.

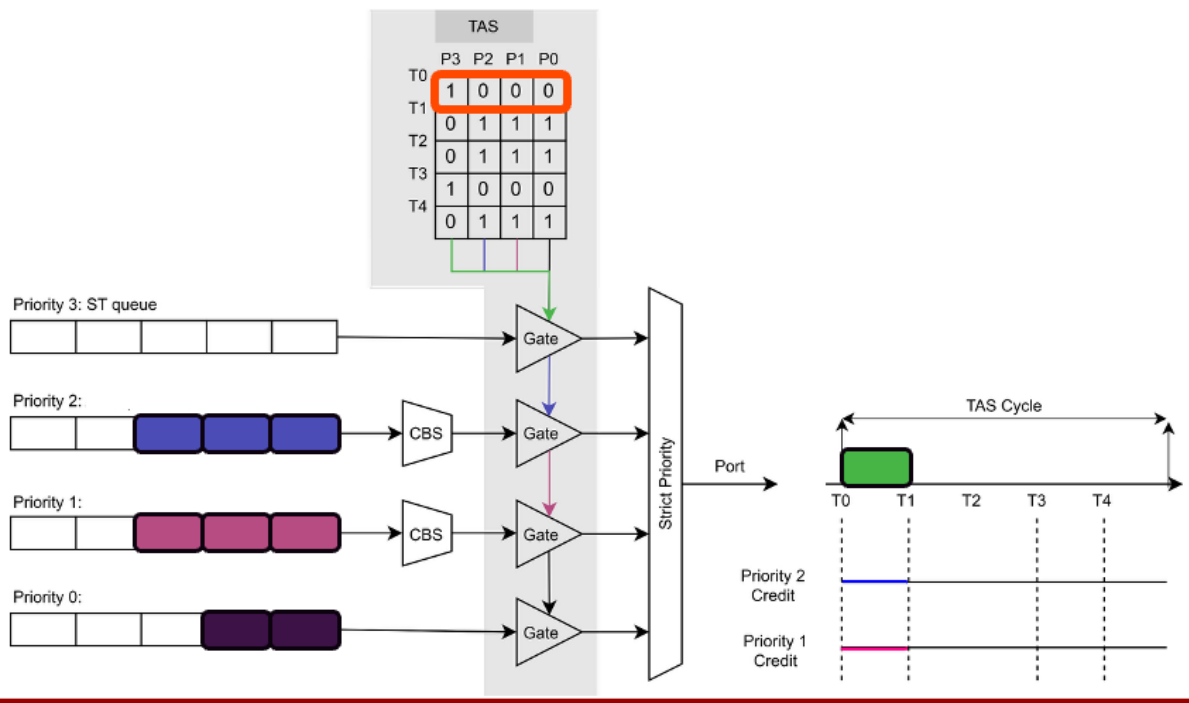


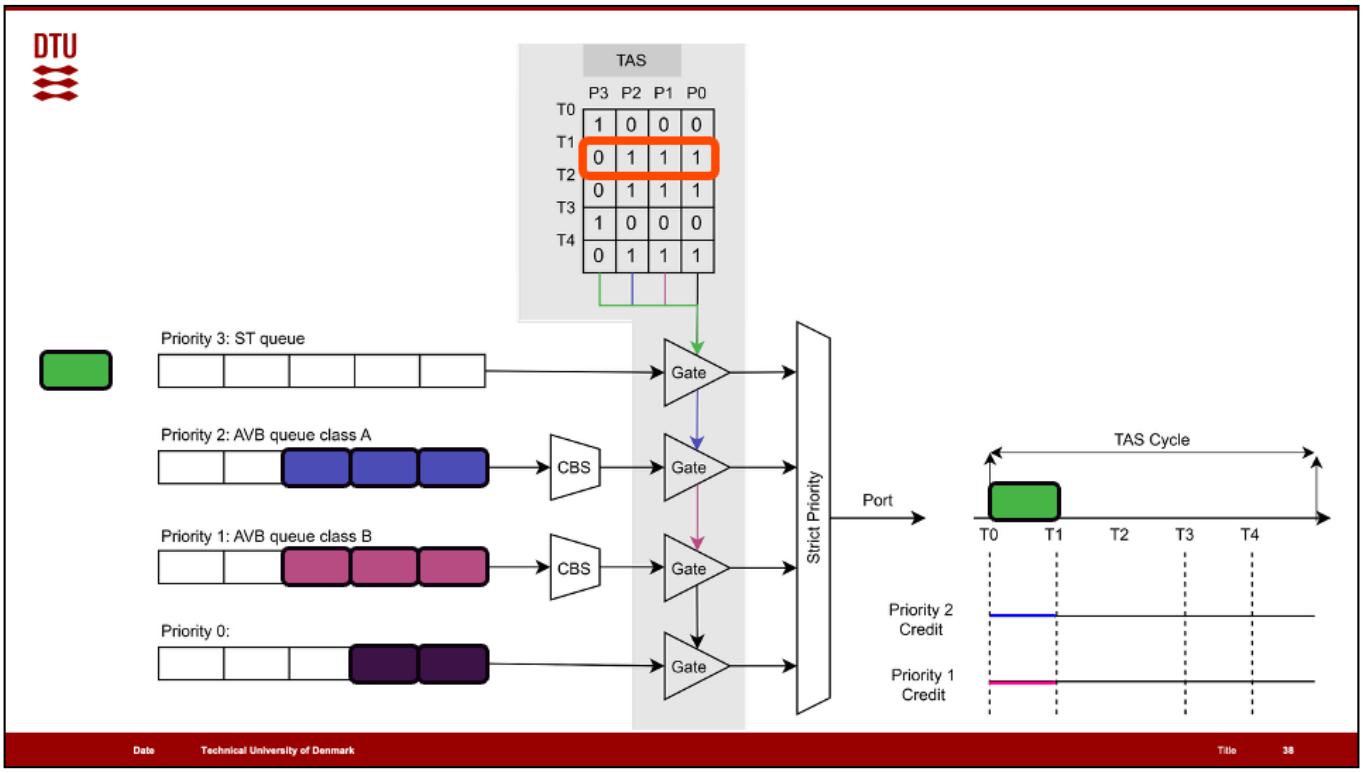
The opening and closing of gates is controlled by the **Gate Control List (GCL)**, a cyclic schedule that specifies which gates must be open or closed during each time interval. This mechanism allows TSN to **close all gates except the one corresponding to the queue that is scheduled to transmit**, thereby preventing interference or blocking from other queues.

As a result, TAS enables the transmission of specific frames at **precisely defined points in time**, providing strong guarantees on both **latency and jitter**, which are essential for highly time-critical applications.

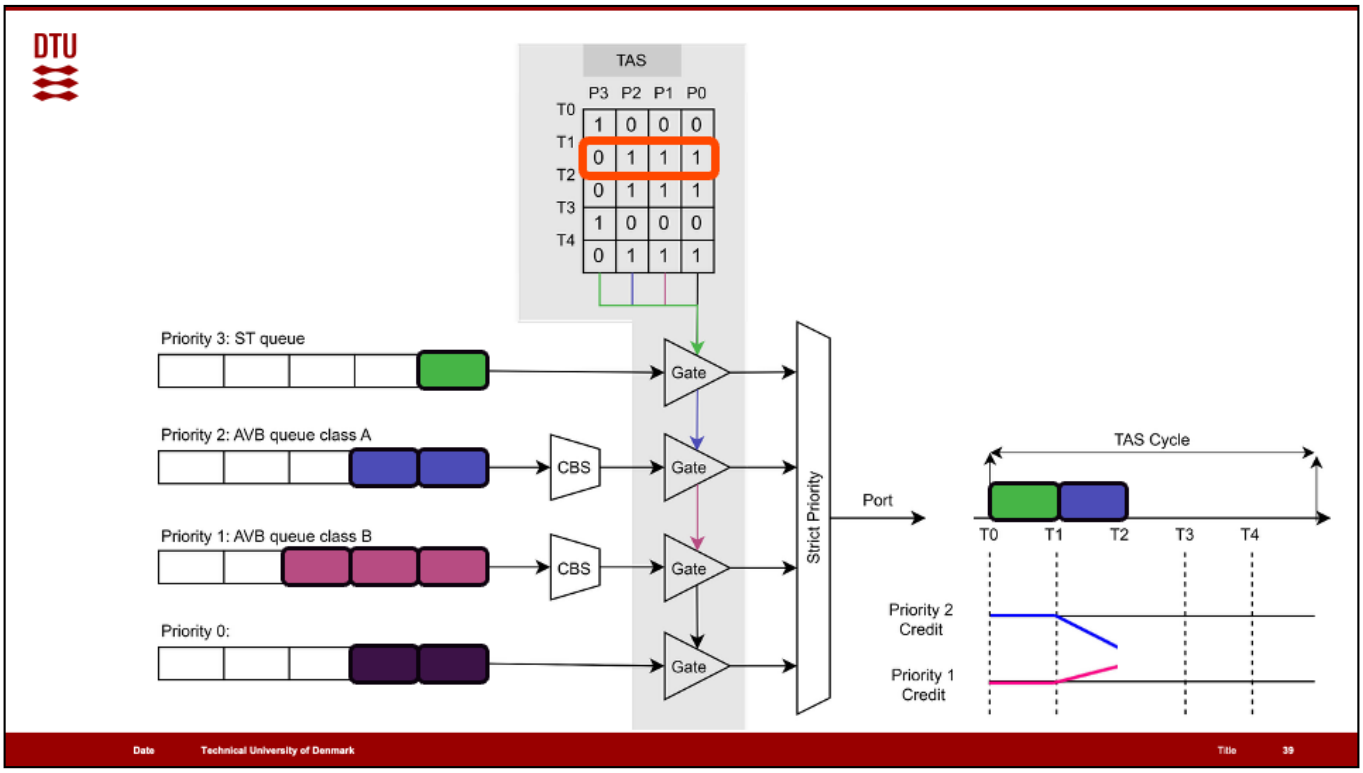


This 3 traffic shapers define 3 types of traffic. First of all, with the highest priority, there is the scheduled traffic. ST traffic follows a fix schedule thanks to the TAS which is typically scheduled offline . As we can see in the example, at T0 only priority 3 gate is open, ensuring an ST transmission without interference and zero jitter.

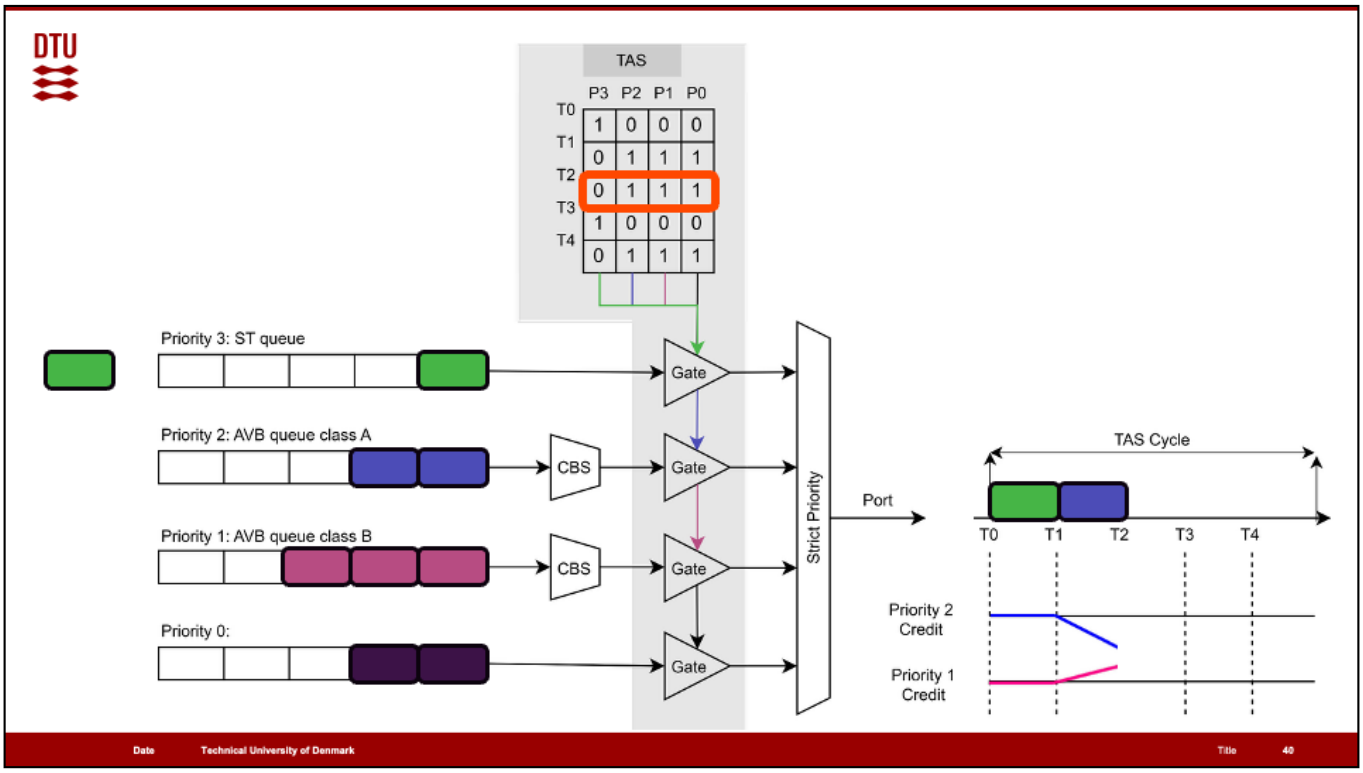

TAS
P3 P2 P1 P0
T0 1 0 0 0
T1 0 1 1 1
T2 0 1 1 1
T3 1 0 0 0
T4 0 1 1 1
Priority 3: ST queue
Priority 2:
Priority 1:
Priority 0:
CBS
CBS
Gate
Gate
Gate
Gate
Strict Priority
Port
TAS Cycle
T0 T1 T2 T3 T4
Priority 2 Credit
Priority 1 Credit



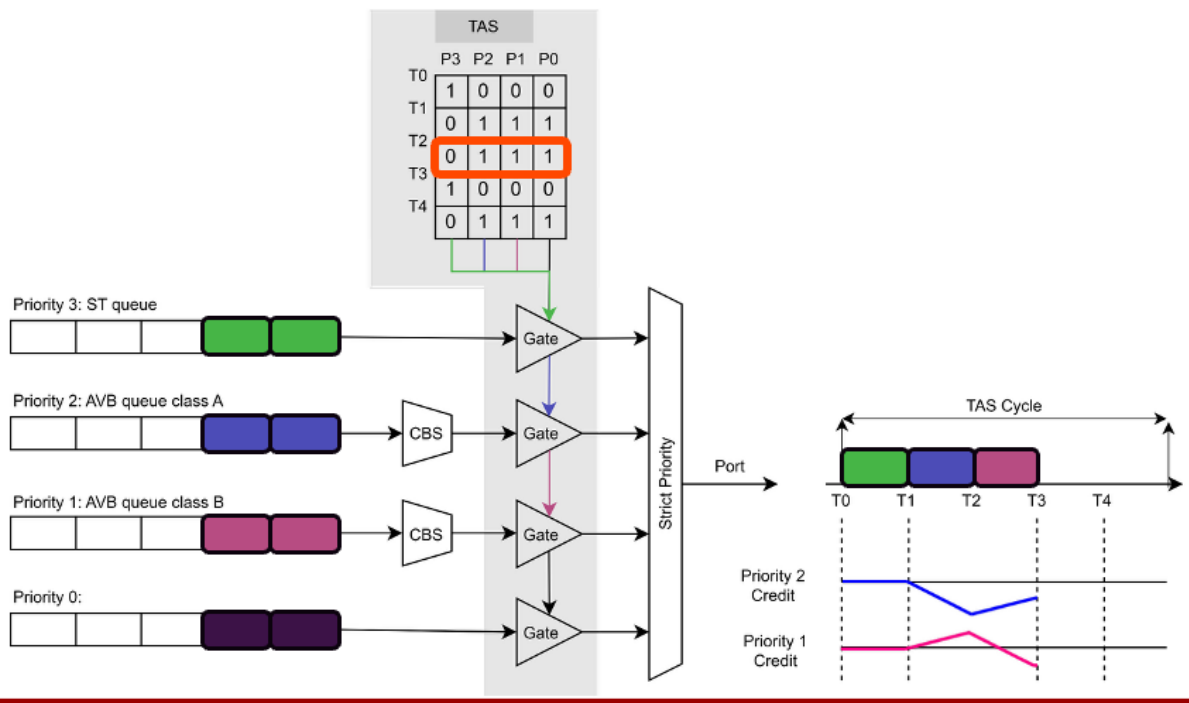
The second TSN type of traffic is AVB traffic. unlike ST which follows a fix schedule, AVB is scheduled online by means of the CBS. As we can see in the example, at T1 ST gate is closed and non-ST gates are open. Since the gates are open and the credit is 0 all 3 non-ST queues are eligible for transmission so the one with the highest priority transmits.

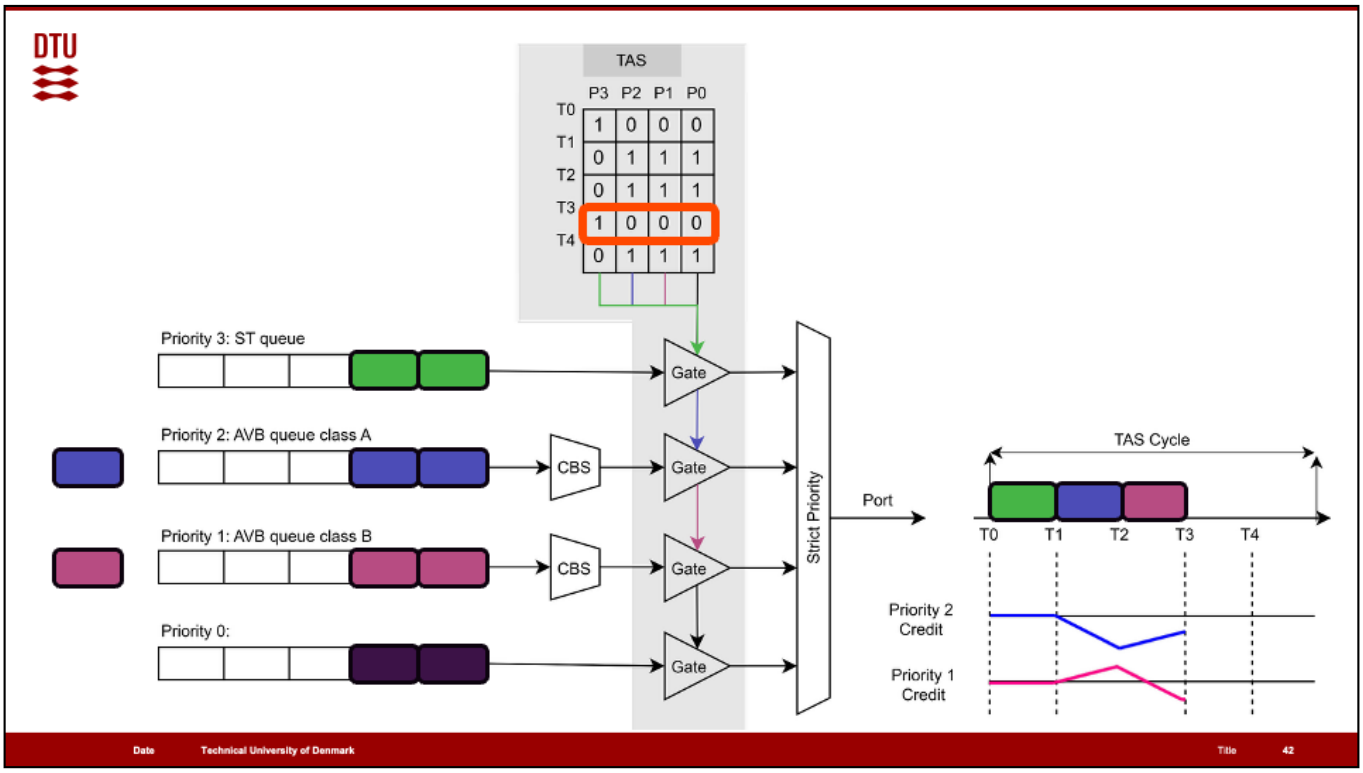


We can also see that since priority 2 queue is transmitting its credit is consumed while priority 1 credit is growing since there are frames waiting to be transmitted.

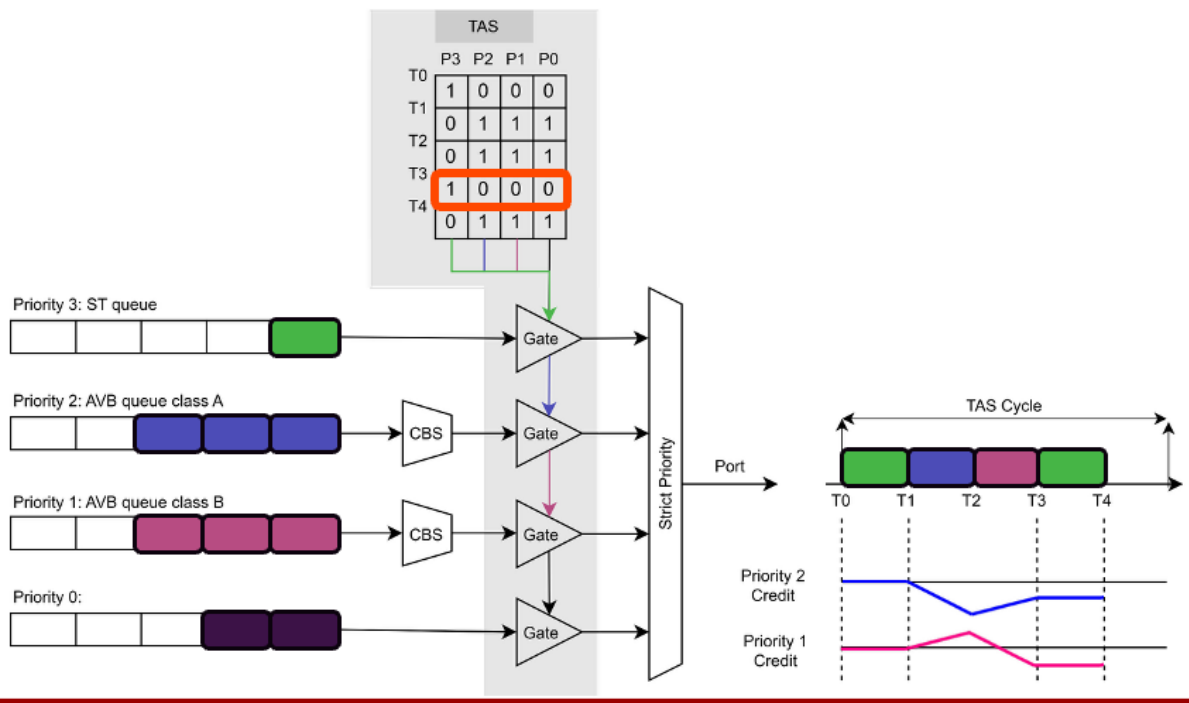


At time 3, since priority 2 credit is negative, an AVB frame with lower priority is transmitted even when there are higher priority frames waiting to be transmitted.


TAS
P3 P2 P1 P0
T0 1 0 0 0
T1 0 1 1 1
T2 0 1 1 1
T3 1 0 0 0
T4 0 1 1 1
Priority 3: ST queue
Priority 2: AVB queue class A
Priority 1: AVB queue class B
Priority 0:
CBS
CBS
Gate
Gate
Gate
Gate
Strict Priority
Port
TAS Cycle
T0 T1 T2 T3 T4
Priority 2 Credit
Priority 1 Credit



At T3 the ST gate opens again and all non-ST gates close to guarantee the transmission of the ST frame as scheduled.


TAS
P3 P2 P1 P0
T0 1 0 0 0
T1 0 1 1 1
T2 0 1 1 1
T3 1 0 0 0
T4 0 1 1 1
Priority 3: ST queue
Priority 2: AVB queue class A
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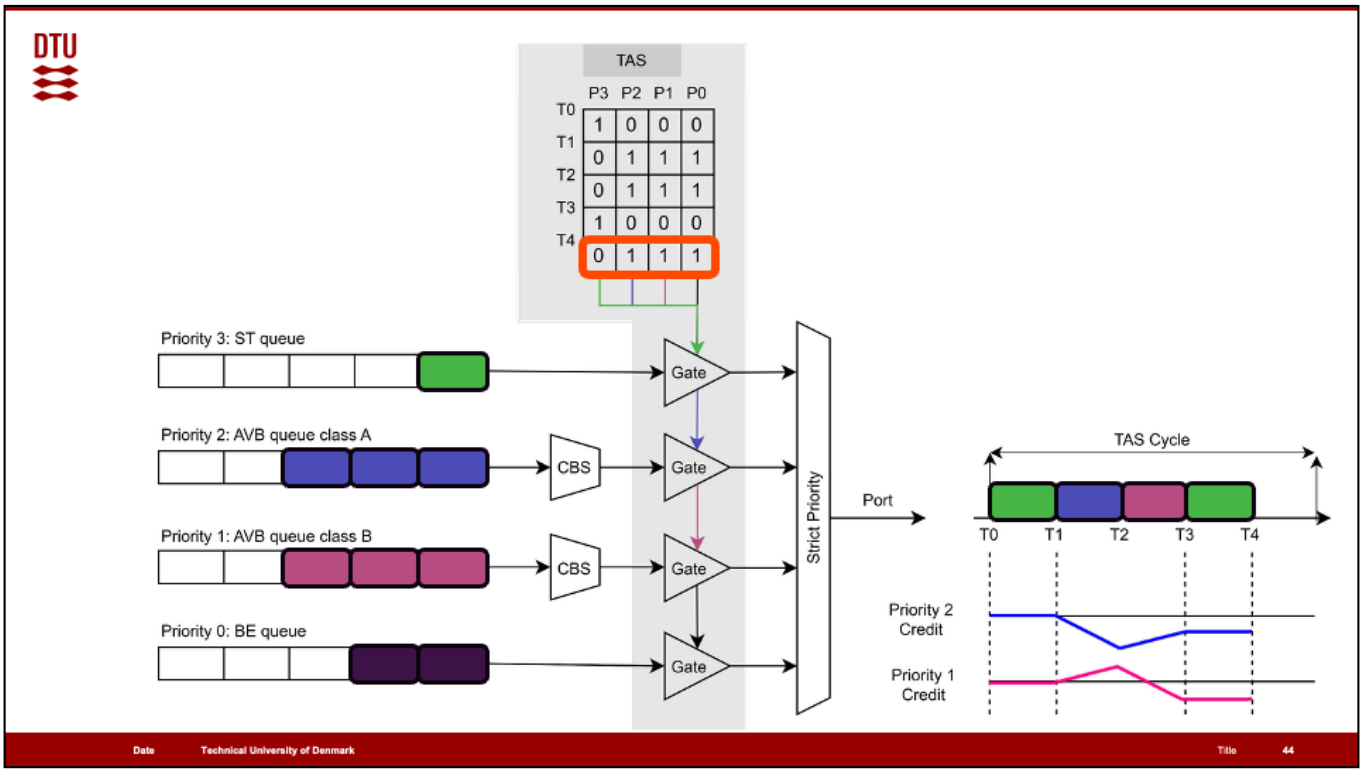
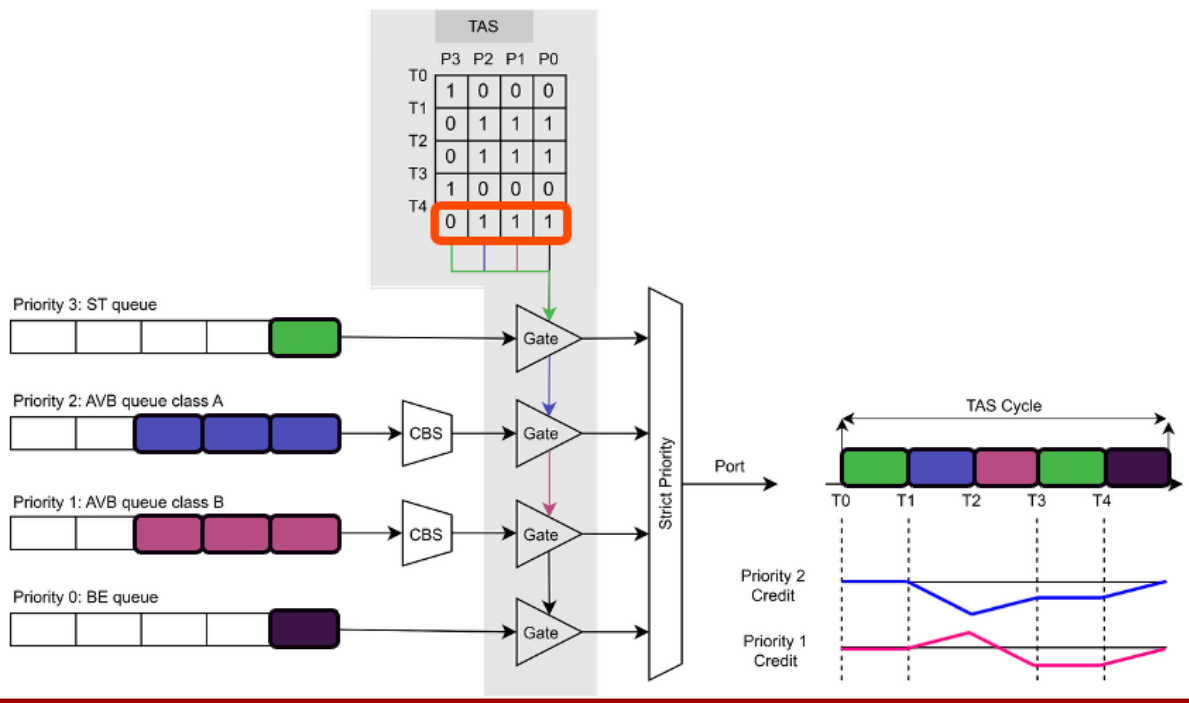


And finally we reach the best effort traffic which can only be transmitted when its gate is open and all higher priority queues are not eligible for transmission or there is no higher priority traffic waiting to be transmitted.


TAS
P3 P2 P1 P0
T0 1 0 0 0
T1 0 1 1 1
T2 0 1 1 1
T3 1 0 0 0
T4 0 1 1 1
Priority 3: ST queue
Priority 2: AVB queue class A
Priority 1: AVB queue class B
Priority 0: BE queue
CBS
CBS
Gate
Gate
Gate
Gate
Strict Priority
Port
TAS Cycle
T0 T1 T2 T3 T4
Priority 2 Credit
Priority 1 Credit

TSN types of traffic

Scheduled Traffic	Audio-Video Bridging (AVB) Traffic	Best Effort (BE) Traffic
+ high determinism + low jitter - low adaptability	+ real-time capabilities + high adaptability - high jitter	+ low impact on higher priority traffic - No real-time guarantees

In summary, TSN defines **three traffic classes**: **Scheduled Traffic (ST)**, **Audio/Video Bridging (AVB)**, and **Best Effort (BE)**. Each of them presents distinct advantages and limitations.

Scheduled Traffic (ST) provides **very high determinism** with **zero jitter**, making it suitable for the most time-critical applications. However, it offers **low adaptability**, since computing an ST schedule is a **computationally expensive task**, as it corresponds to an **NP-hard problem**. In addition, TSN provides **limited support for online TAS reconfiguration**, which further reduces flexibility.

Audio/Video Bridging (AVB) offers **real-time capabilities** by enforcing **bounded delays** and is **highly adaptable**. It can efficiently handle **periodic and aperiodic traffic** thanks to its **online scheduling**, as well as the **dynamic addition or removal of traffic flows**. Nevertheless, AVB **cannot guarantee very low jitter**, which limits its suitability for the most stringent real-time requirements.

Best Effort (BE) traffic has a **low impact on higher-priority traffic classes**, but it **does not provide any real-time guarantees** in terms of

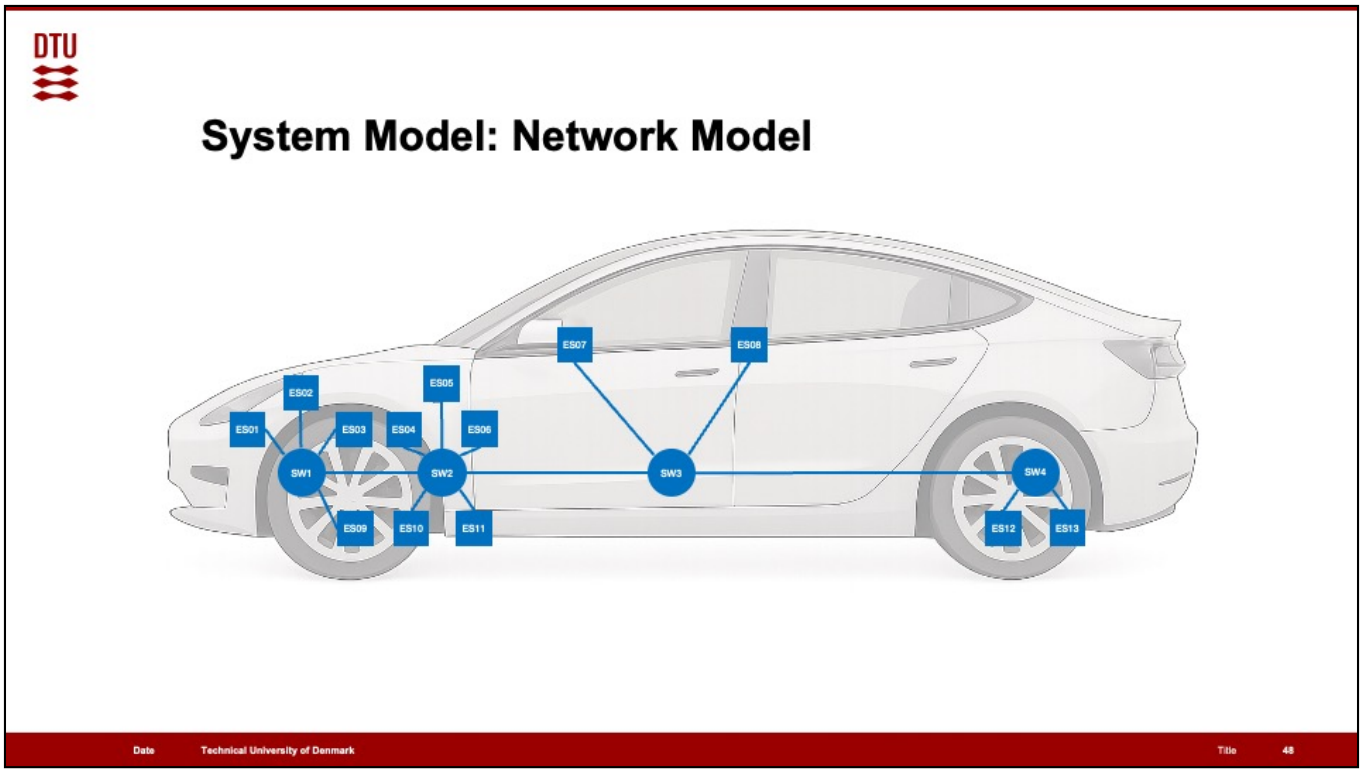
delay or jitter.

Together, these three traffic classes allow TSN to balance **determinism, flexibility, and efficiency**, depending on the requirements of the application.

Agenda

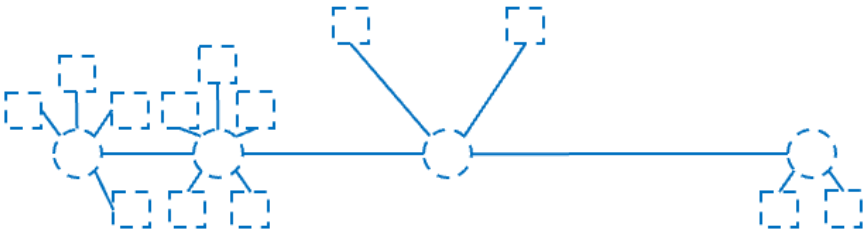
- System model
  - Network model
  - Traffic model
- Analysis

In the following slides, we will introduce the **network and traffic model** that will be used as the basis for our subsequent **analysis of TSN**.



Regarding the **network model**, it is sufficient to focus on the **communication links**, taking into account that links between network nodes are **full-duplex**. Therefore, each full-duplex link is modeled as **two unidirectional links with opposite directions**.

System Model: Network Model



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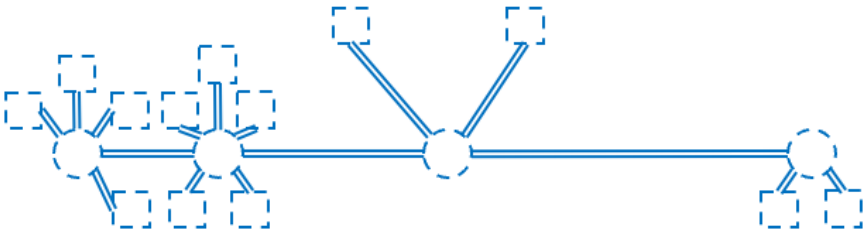

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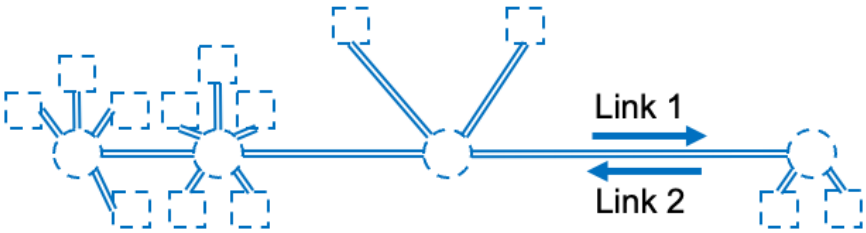

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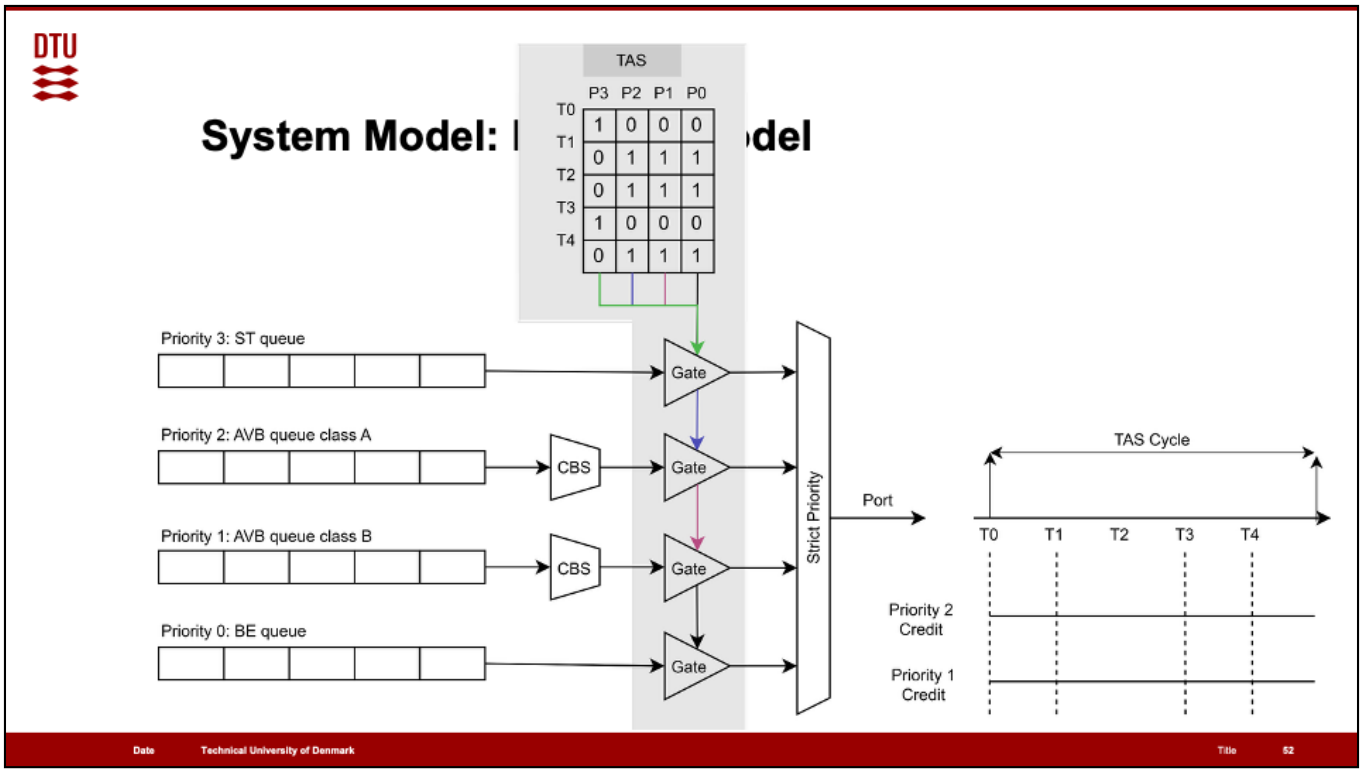


On the other hand, in this presentation—as well as in **Mini-Project 2**—we will analyze a **simplified version of TSN** that **does not include the TAS** and supports **only three priority levels**. The **two highest priorities** are dedicated to **AVB traffic** and therefore implement the **CBS**.

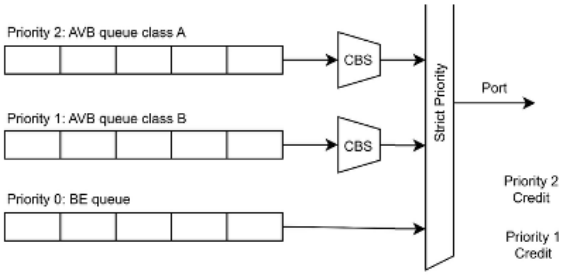
System Model: Network Model

For all links $\forall\, l_i \in \mathbb{L}$

Define Bandwidth BW_i

And for every AVB queue in the link $\forall\, P_j^i \in AVB$

Define sending and idle slope $\alpha_j^{i^-}$ $\alpha_j^{i^+}$



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Under this model, for **each link in the network**, we consider the **output port that feeds it**. As a result, each link is characterized by its **bandwidth**. In addition, for each **AVB queue** at the corresponding output port, we define the **rates at which credit is consumed and replenished**, known as **sendSlope** and **idleSlope**, respectively.

System Model: Network Model

$$\alpha_j^{i^-} + \alpha_j^{i^+} = 1$$

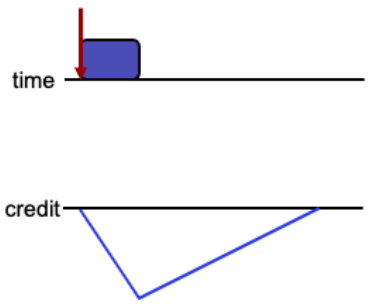
$\alpha_j^{i^+} \times 100 =$ Percentage of reserved bandwidth(RBW[%])

Although there are multiple ways to define **sendSlope** and **idleSlope**, one of the most common approaches is to choose them so that **they sum to 1**. With this convention, the value of **idleSlope** directly corresponds to the **fraction of link bandwidth reserved** for that AVB priority on that link.

System Model: Network Model

$$\alpha_j^{i^-} + \alpha_j^{i^+} = 1$$

$\alpha_j^{i^+} \times 100 =$ Percentage of reserved bandwidth(RBW[%])



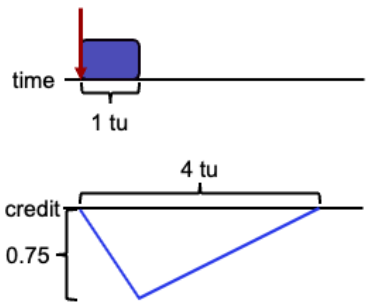
$\alpha_j^{i^-} = 0.75 \rightarrow \alpha_j^{i^+} = 0.25$

EXAMPLE

System Model: Network Model

$$\alpha_j^{i^-} + \alpha_j^{i^+} = 1$$

$\alpha_j^{i^+} \times 100 =$ Percentage of reserved bandwidth(RBW[%])



$$\alpha_j^{i^-} = 0.75 \rightarrow \alpha_j^{i^+} = 0.25 \rightarrow RBW[\%] = \frac{1\,tu}{4\,tu} = 0.25$$

EXAMPLE

System Model: Network Model

$$\alpha_j^{i^-} + \alpha_j^{i^+} = 1$$

$\alpha_j^{i^+} \times 100 =$ Percentage of reserved bandwidth(RBW[%])

The following analysis is valid for P_j on link l only if:

$$\sum_{\forall P_i | P_i \geq P_j} RBW_i^l \leq 1$$

Therefore, I recommend computing the **idleSlope** for P_j on link l as follows:

$$\alpha_j^{i^+} = \frac{U_j^l}{\sum_{\forall P_i \in AVB} U_i^l} \times (1 - U_{BE}^l)$$

However, it is important to note that the **analysis presented next** is only valid when the **sum of the bandwidth reserved by the priority under analysis and all higher priorities is less than or equal to 1**.

For this reason, I recommend distributing the **remaining bandwidth**, after subtracting the utilization of **Best Effort (BE)** traffic, **proportionally to the utilization of the AVB priorities**. This weighted allocation ensures that the assumptions required for the analysis are satisfied while preserving fairness among AVB traffic classes.

System Model: Traffic Model

$\Gamma = \{m_i\ (C_i, T_i, D_i, P_i, \mathcal{L}_i) \mid i = 1 \dots N\}$

- C_i → Transmission time $(frame.size/BWD)$
- T_i → Period
- D_i → Deadline
- P_i → Priority
- $\mathcal{L}_i = \{\mathcal{L}_i(0) = l_i, \dots, \mathcal{L}_i(n-1) = l_k\}$ → path
  - n → number of links in the path
  - $l_i, l_k \in \mathbb{L}$ → links in path

On the other hand, traffic is modeled as **flows**, referred to as **streams**. A stream represents a sequence of frames that share similar characteristics, such as **frame size**, **periodicity**, and other traffic parameters.

Streams can be seen as the **communication equivalent of tasks** in real-time systems, where **each frame corresponds to a job or task instance** of the stream.

Streams are characterized by several parameters. First, their **transmission time**, which is computed by dividing the **frame size** by the **link bandwidth**. In addition, each stream is defined by its **period**, **deadline**, and **priority**.

Finally, streams are associated with a **path**, which represents the **set of links in the network model** that the stream must traverse in order to travel from the **source end-system to the destination end-system**.

Agenda

- System model
  - Network model
  - Traffic model
- **Analysis:**
  - **Worst-Case Response Time Analysis (WCRTA)** based on **eligible interval**

The analysis presented next is a **Worst-Case Response Time (WCRT) analysis** based on the concept of **eligible intervals**. More detailed explanations, as well as formal proofs of this analysis, can be found in the paper referenced in **Mini-Project 2**.

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} WCRT_i^l$$

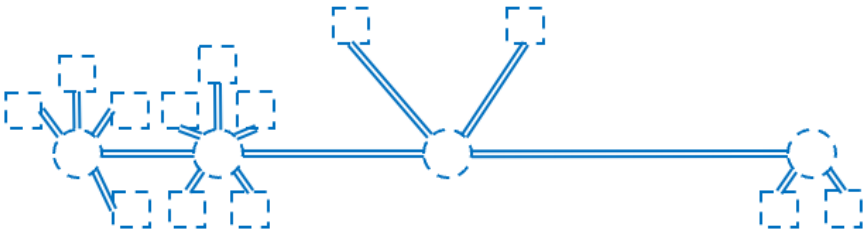
First, this analysis is **compositional**, meaning that it computes the **Worst-Case Response Time (WCRT)** from the **source to the destination** by **summing the WCRT contributions of each link** along the stream's path.

Other parameters, such as **switch delay**—that is, the time elapsed between a frame being received at an input port and being enqueued at the corresponding output port—or **propagation delay**, namely the time required for a frame to travel across a link, are assumed to be **negligible** when compared to the **transmission time**. The transmission time is defined as the interval between the moment a frame enters the transmission queue and the moment it is fully transmitted onto the link.

When necessary, these additional delays can be incorporated into the analysis as **constant offsets**, depending on the specific characteristics of the switch or the link.

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} WCRT_i^l$$



EXAMPLE

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EXAMPLE

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} \boxed{WCRT_i^l}$$

$$WCRT_i^l = SPI_i^l + HPI_i^l + LPI_i^l + C_i$$

On the other hand, the **Worst-Case Response Time (WCRT)**—or, in this context, the **transmission time**—is computed by taking into account **three types of interference**, in addition to the transmission time of the frame under analysis itself.

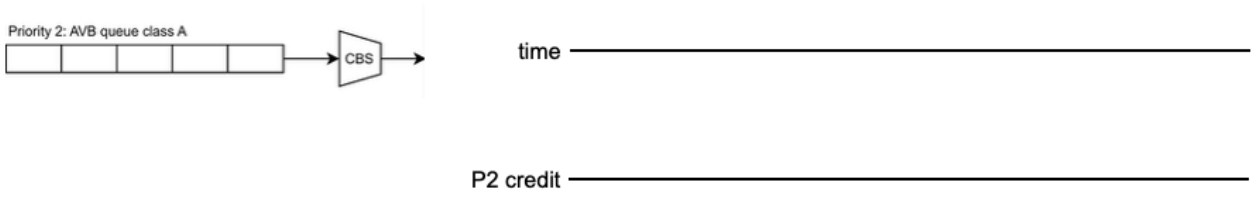
Same-Priority Interference (SPI): interference caused by frames with the **same priority** as the frame under analysis.

Higher-Priority Interference (HPI): interference caused by frames with **higher priority** than the frame under analysis.

Lower-Priority Interference (LPI): **blocking** caused by frames with **lower priority**, which may delay the transmission of the frame under analysis.

Analysis

$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$

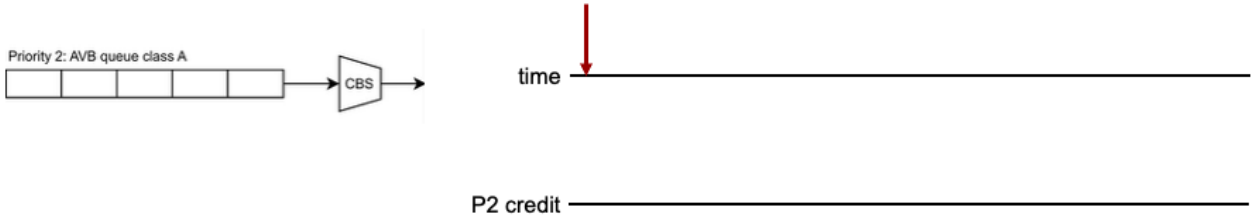


Let us begin by analyzing the **Same-Priority Interference (SPI)** for a **single AVB queue**.

Analysis

$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$

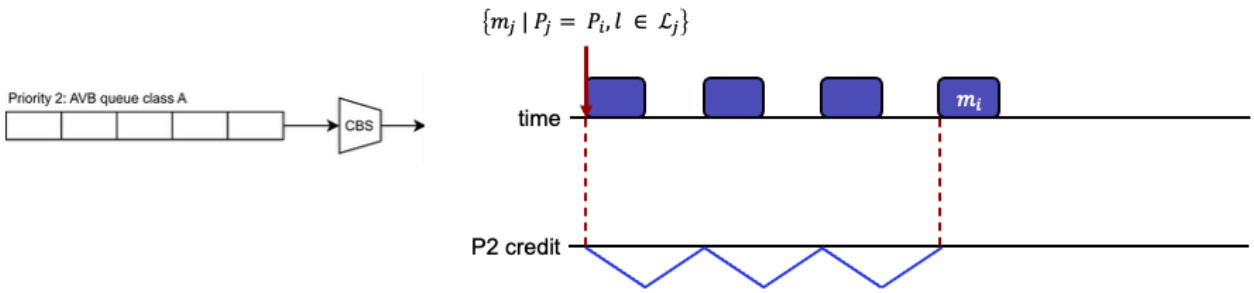
$\{m_j \mid P_j = P_i, l \in \mathcal{L}_j\}$



When a frame arrives at the **transmission queue**, if the **deadline of the streams is less than or equal to their period**, then at most **one frame per stream** with the **same priority** as the frame under analysis can be present in the queue. Otherwise, one of the frames already in the queue would have **missed its deadline**.

Analysis

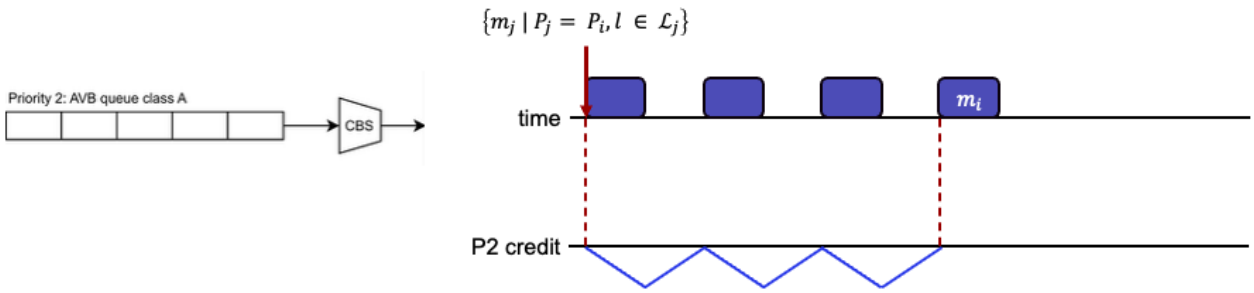
$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$



Therefore, for the frame under analysis to be transmitted, **all frames ahead of it in the queue** must be transmitted first. However, it is not sufficient to simply sum the **transmission times of the same-priority frames**. It is also necessary to account for the **time required to recover the credit** consumed by each of those transmissions.

Analysis

$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$



$$SPI_i^l = \sum_{\{m_j \mid m_j \neq m_i, P_j = P_i, l \in \mathcal{L}_j\}} C_j \times \left(1 + \frac{\alpha_i^{l^-}}{\alpha_i^{l^+}}\right)$$

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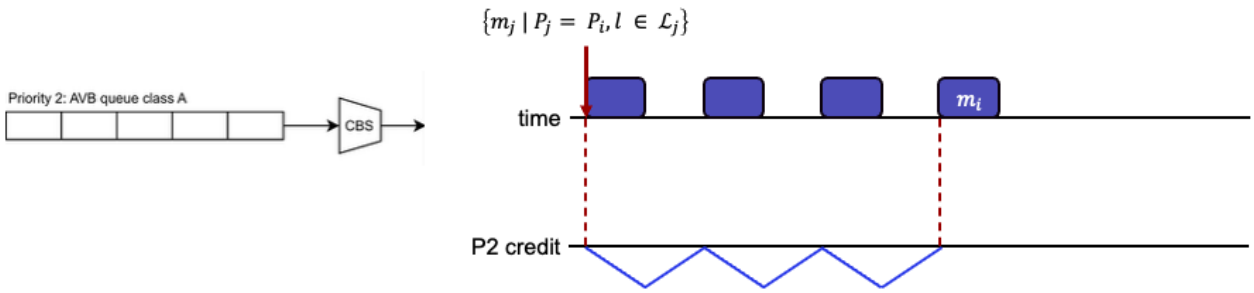

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Next, we will compute the **WCRT components corresponding to higher- and lower-priority interference**. Before doing so, it is important to note the following observation.

If the **credit level is the same at the beginning and at the end of the analysis interval**, then it is irrelevant whether interference from other sources has occurred or not. This is because any such interference is effectively reflected as **credit recovery time**, which has already been accounted for in the **Same-Priority Interference (SPI)** term.

Consequently, this interference **must not be added again**, as doing so would result in double counting.

Analysis

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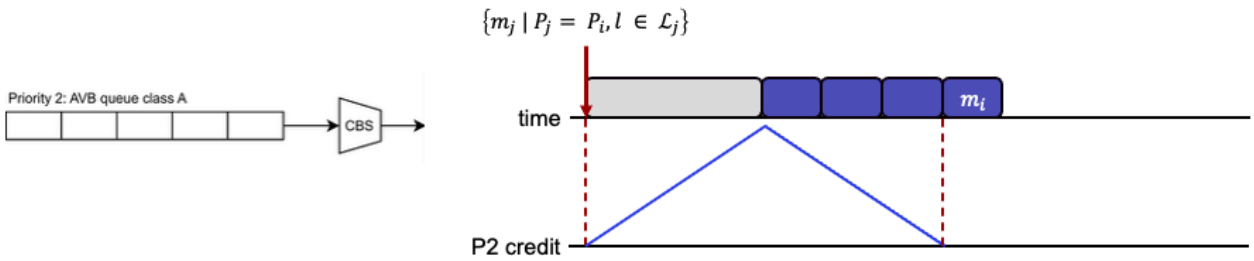

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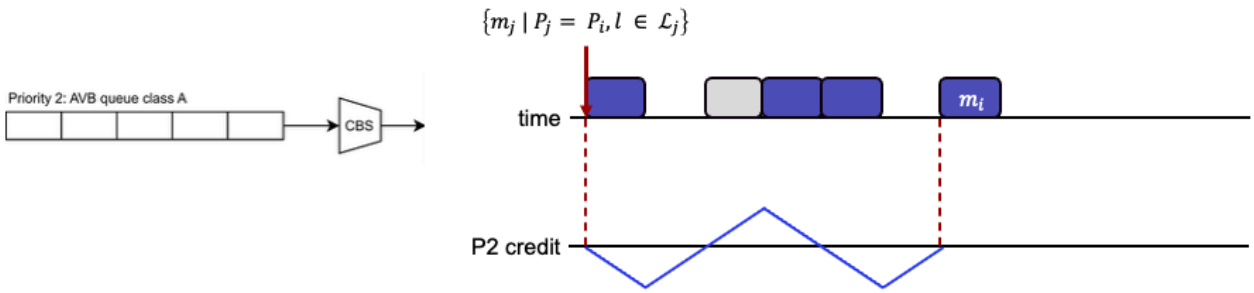

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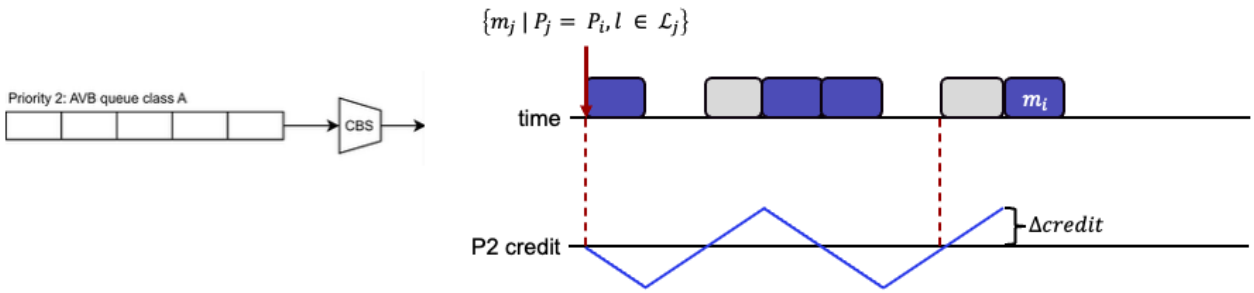
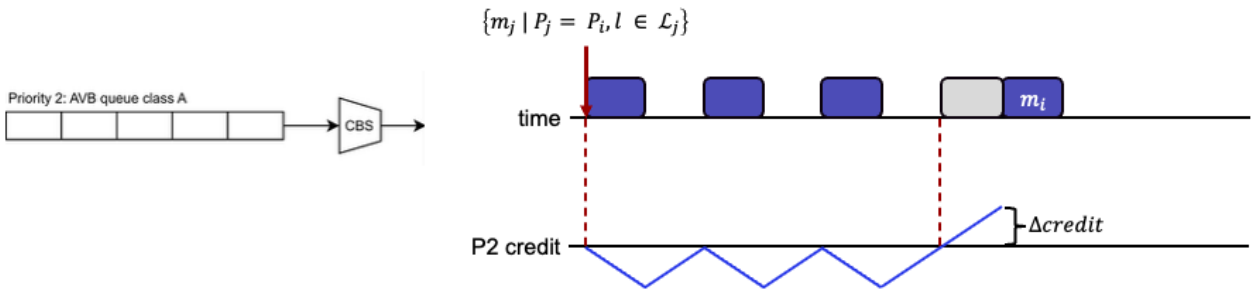


In this sense, any **additional delay beyond the Same-Priority Interference (SPI)** will be reflected as a **positive difference in credit** between the beginning and the end of the analysis interval.

Therefore, if we can determine the **maximum credit that can be accumulated for each priority**, we can compute the **Higher- and Lower-Priority Interference** by calculating the **time required to reach that credit level**.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$



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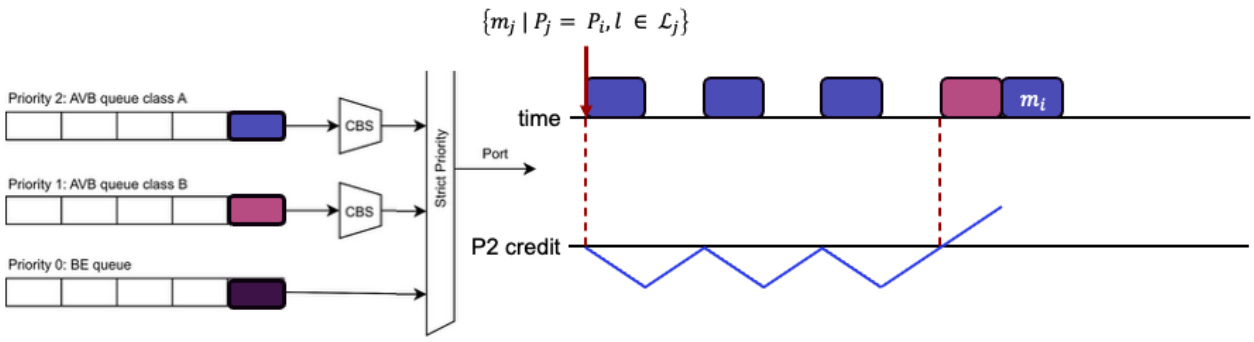
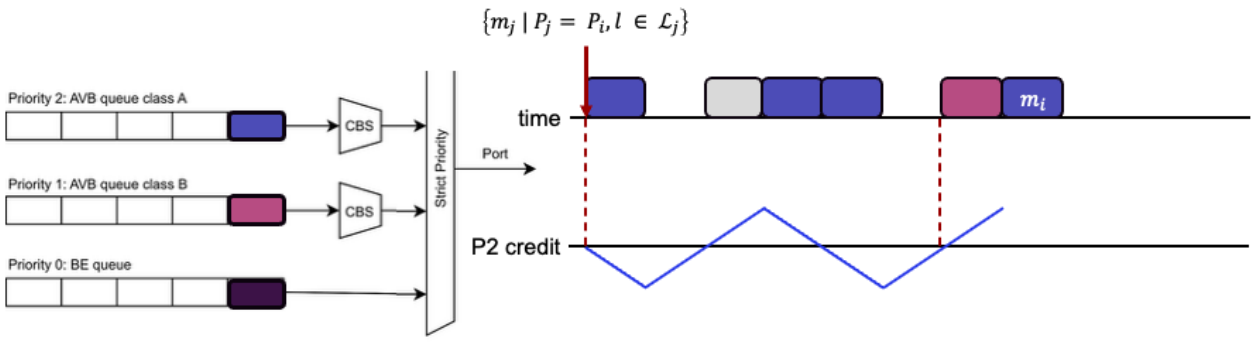


Let us start by analyzing the **highest-priority AVB queue or class A**. On the one hand, this queue can only be **blocked by lower-priority traffic**, since there is **no higher-priority traffic** that could interfere with it. Moreover, blocking can occur **only when its credit is negative**, because otherwise this queue would win access to the link due to its highest priority.

As a result, the **maximum credit** is reached when the queue has **negative credit**, and **just before the credit recovers to zero**, a **lower-priority frame with the largest transmission time** starts transmitting.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

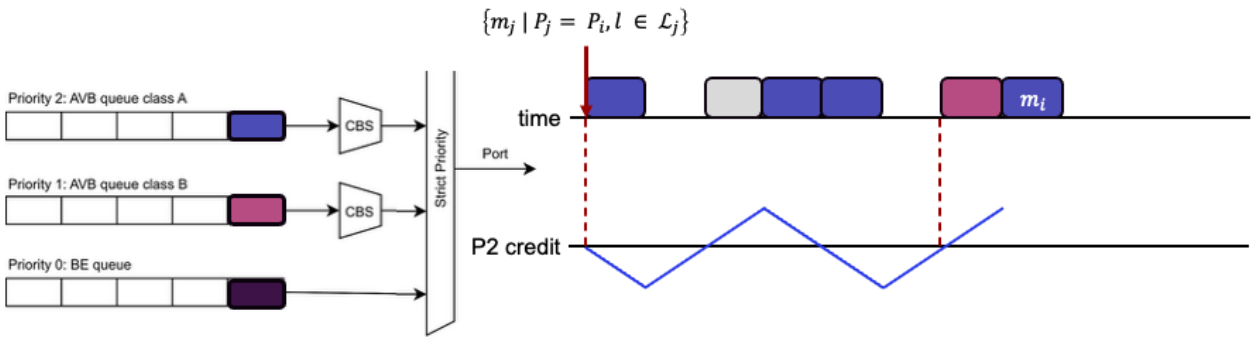


Let us start by analyzing the **highest-priority AVB queue or class A**. On the one hand, this queue can only be **blocked by lower-priority traffic**, since there is **no higher-priority traffic** that could interfere with it. Moreover, blocking can occur **only when its credit is negative**, because otherwise this queue would win access to the link due to its highest priority.

As a result, the **maximum credit** is reached when the queue has **negative credit**, and **just before the credit recovers to zero**, a **lower-priority frame with the largest transmission time** starts transmitting.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{\cancel{HPI_i^l}^{\,0} + \cancel{LPI_i^l}^{\,\max_{m_j \mid P_j < P_i, l \in \mathcal{L}_j} C_j}} + C_i$$

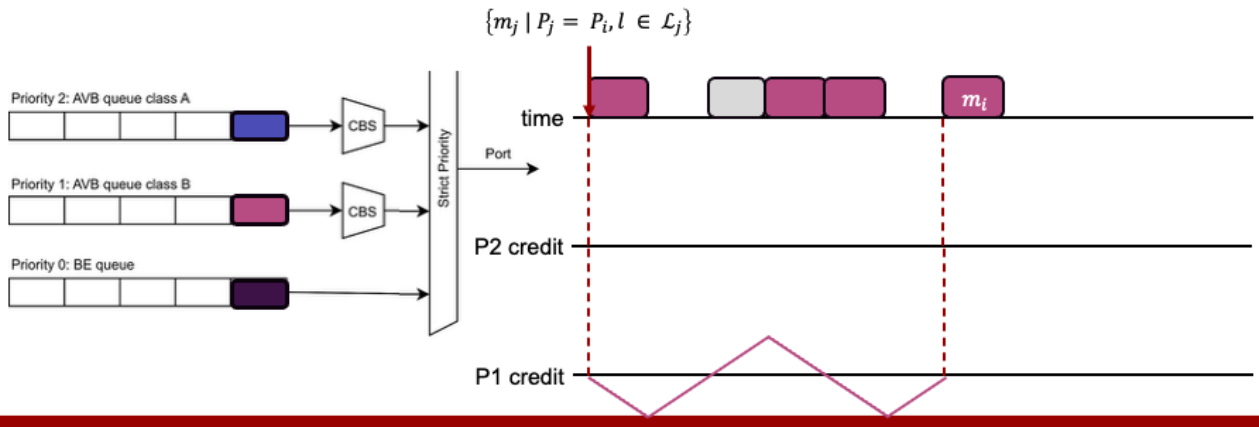


Under these conditions:

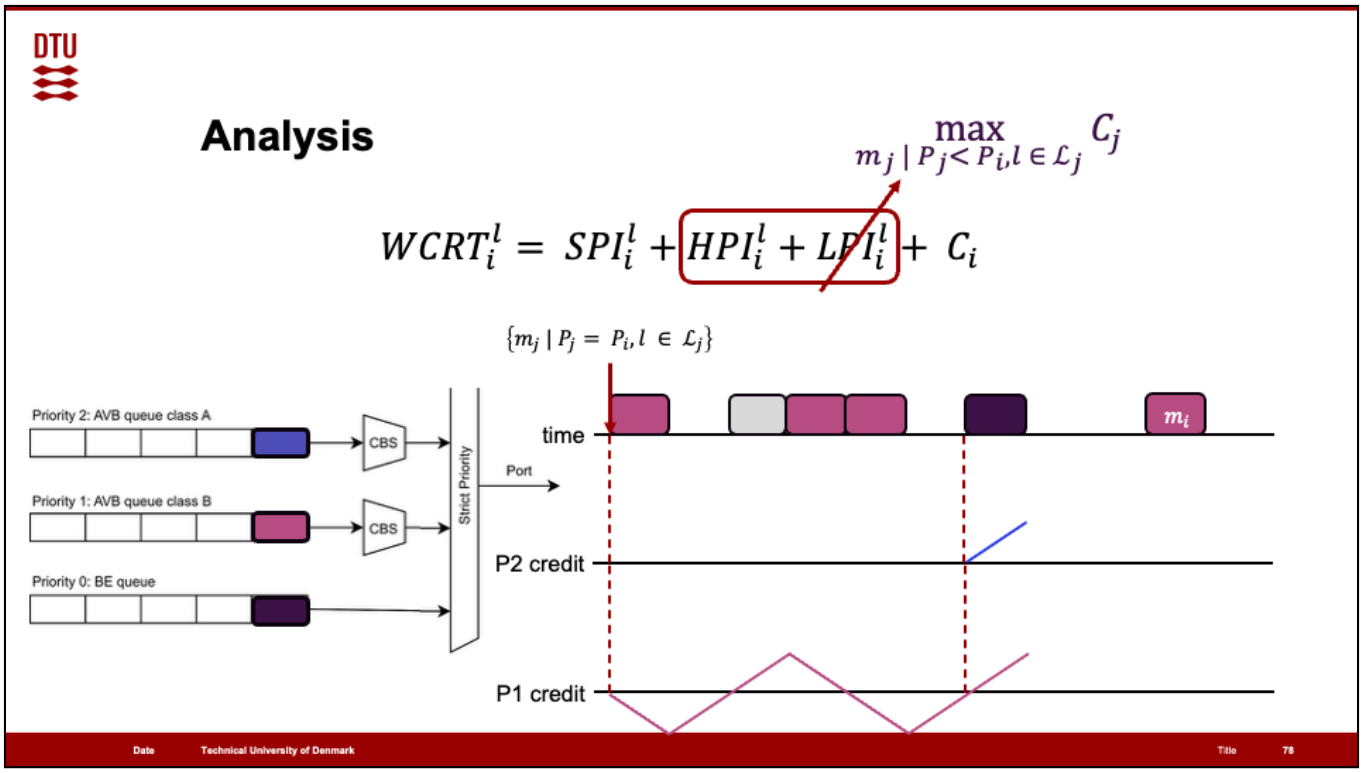
- **Higher-Priority Interference (HPI) = 0**, since no higher-priority traffic exists.
- **Lower-Priority Interference (LPI)** is equal to the **maximum transmission time among all lower-priority streams**.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$



For **AVB Class B traffic**, the **Lower-Priority Interference (LPI)** is **the same as for AVB Class A**, under the same conditions described previously.

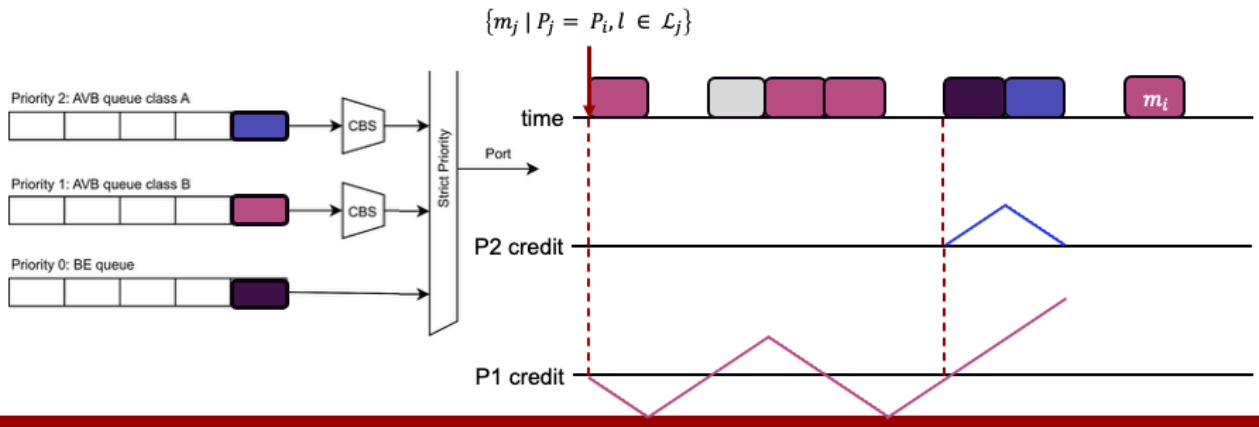


However, in this case, an additional source of interference must be considered. During the blocking caused by lower-priority traffic, the **credit of the higher-priority AVB Class A queue also increases**. As a consequence, in addition to LPI, we must now account for **Higher-Priority Interference (HPI)**.

Analysis

$$LPI_i^l \times \frac{\alpha_H^{l\,+}}{\alpha_H^{l\,-}} \qquad \max_{m_j \mid P_j < P_i, l \in \mathcal{L}_j} C_j$$

$$WCRT_i^l = SPI_i^l + \boxed{\cancel{HPI_i^l} + \cancel{LPI_i^l}} + C_i$$

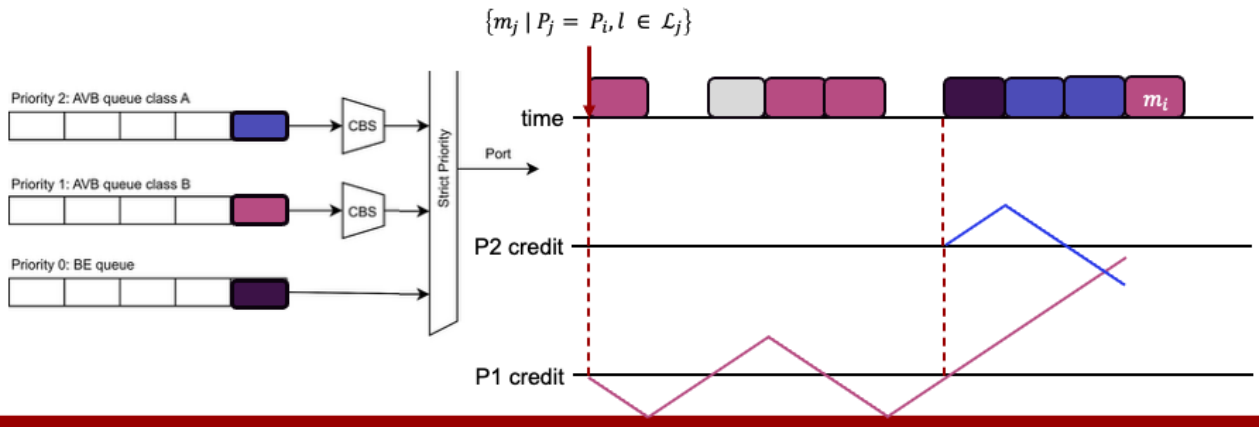


In the **worst-case scenario**, the accumulated **Class A credit**—which built up during the blocking interval—is **fully consumed first**.

Analysis

$$LPI_i^l \times \frac{\alpha_H^{l\,+}}{\alpha_H^{l\,-}} + \max_{m_j \mid P_j > P_i, l \in \mathcal{L}_j} C_j$$

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$



Then, **just when the Class A credit reaches zero**, the transmission of the **largest frame among the higher-priority (Class A) streams** begins.

Therefore, for AVB Class B traffic:

- **LPI** is identical to that of AVB Class A.
- **HPI** consists of the time required to consume the accumulated **Class A credit**, plus the **maximum transmission time** of the higher-priority streams.

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} WCRT_i^l$$

$$WCRT_i^l = SPI_i^l + HPI_i^l + LPI_i^l + C_i$$

$$SPI_i^l = \sum_{\{m_j \mid m_j \neq m_i, P_j = P_i, l \in \mathcal{L}_j\}} C_j \times \left(1 + \frac{\alpha_i^{l^-}}{\alpha_i^{l^+}}\right)$$

$$LPI_i^l = \max_{m_j \mid P_j < P_i, l \in \mathcal{L}_j} C_j$$

AVB Class A

$$HPI_i^l = 0$$

AVB Class B

$$HPI_i^l = LPI_i^l \times \frac{\alpha_H^{l^+}}{\alpha_H^{l^-}} + \max_{m_j \mid P_j > P_i, l \in \mathcal{L}_j} C_j$$

Finally, this slide **summarizes the analytical formulas** used in the **Worst-Case Response Time analysis for AVB traffic**.

DTU